

Marine Microbial Biodiversity

Lecture 03

Microbial Metabolism Pt.1

Matteo Selci, PhD

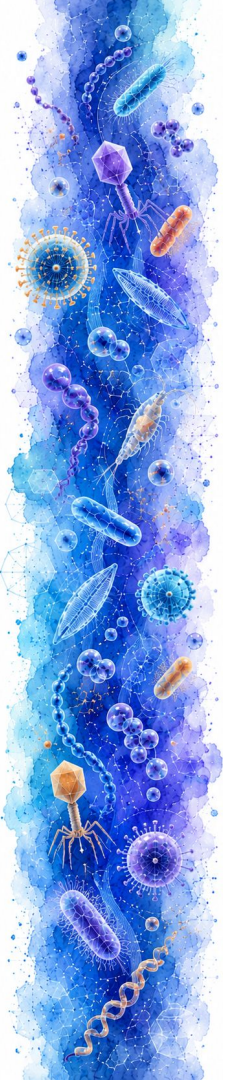
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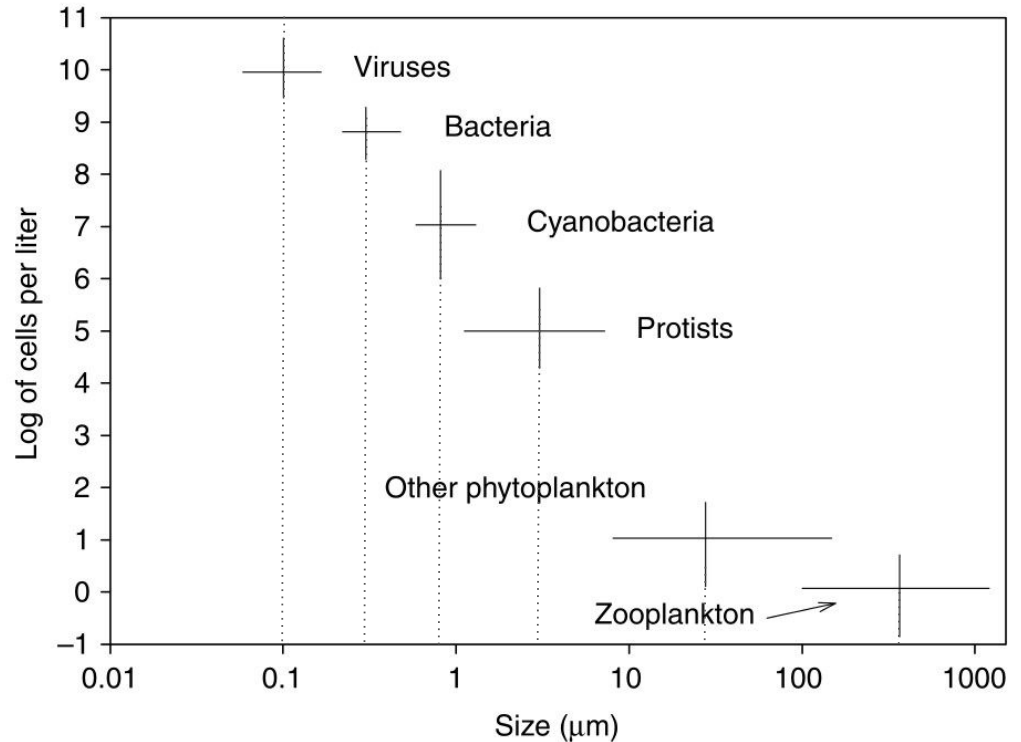


Link for the Marine Microbial Biodiversity slides

<https://github.com/MatS792/Marine-Microbiol-Biodiversity>

Homeworks On Abundance And Size

If we have 10,000,000,000 billion cells in 1 L of seawater, how many cells do we have in 1 mL?



Homeworks On Abundance And Size

If we have 10000000000 billion cells in 1 L of seawater, how many cells do we have in 1 mL?

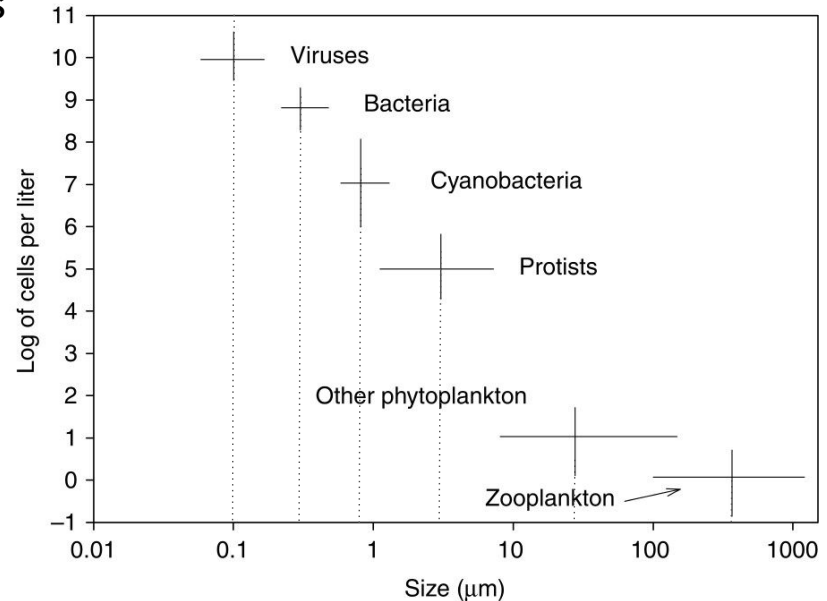
10000000000 billion cells = 10^{10} cells

1 L = 1000 mL

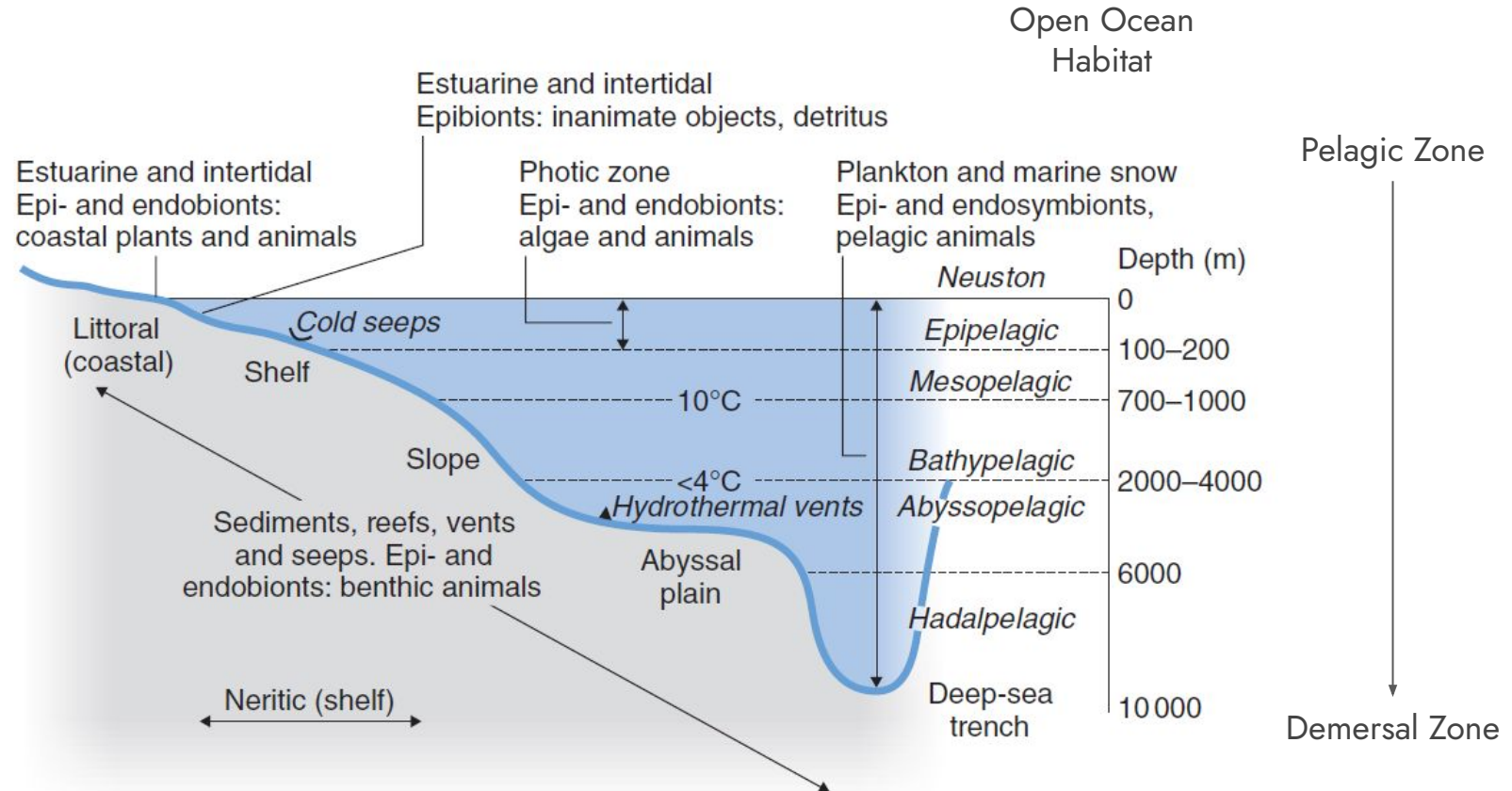
1×10^{10} cells : 1000 mL = X : 1 mL

$X = (1 \times 10^{10} \text{ cells} \times 1 \text{ mL}) / 1000 \text{ mL}$

$X = 1 \times 10^7 \text{ cells} = 10 \text{ million cells}$



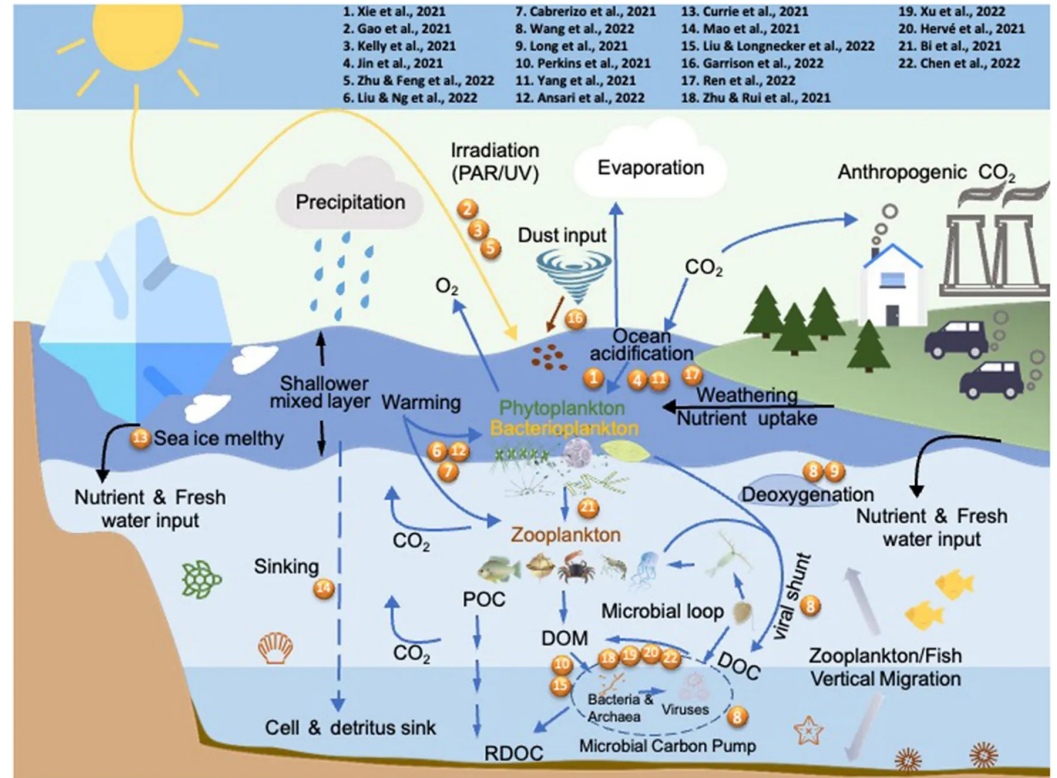
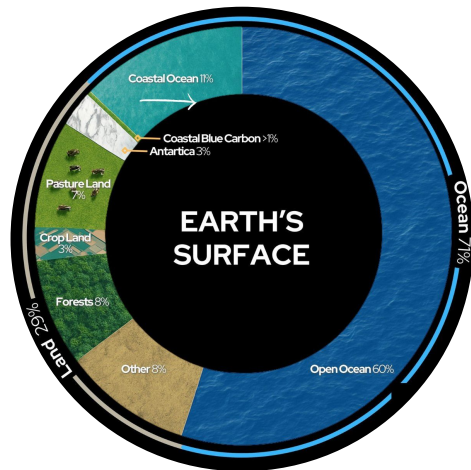
Environmental Parameters VS Microbial Lifestyles



The major ecological zones of the oceans and marine microbial habitats

Environmental Parameters Vs Microbial Lifestyles

By controlling a large portion of biogeochemical cycles, microbes significantly influence element cycling on our planet, with direct consequences on environmental parameters



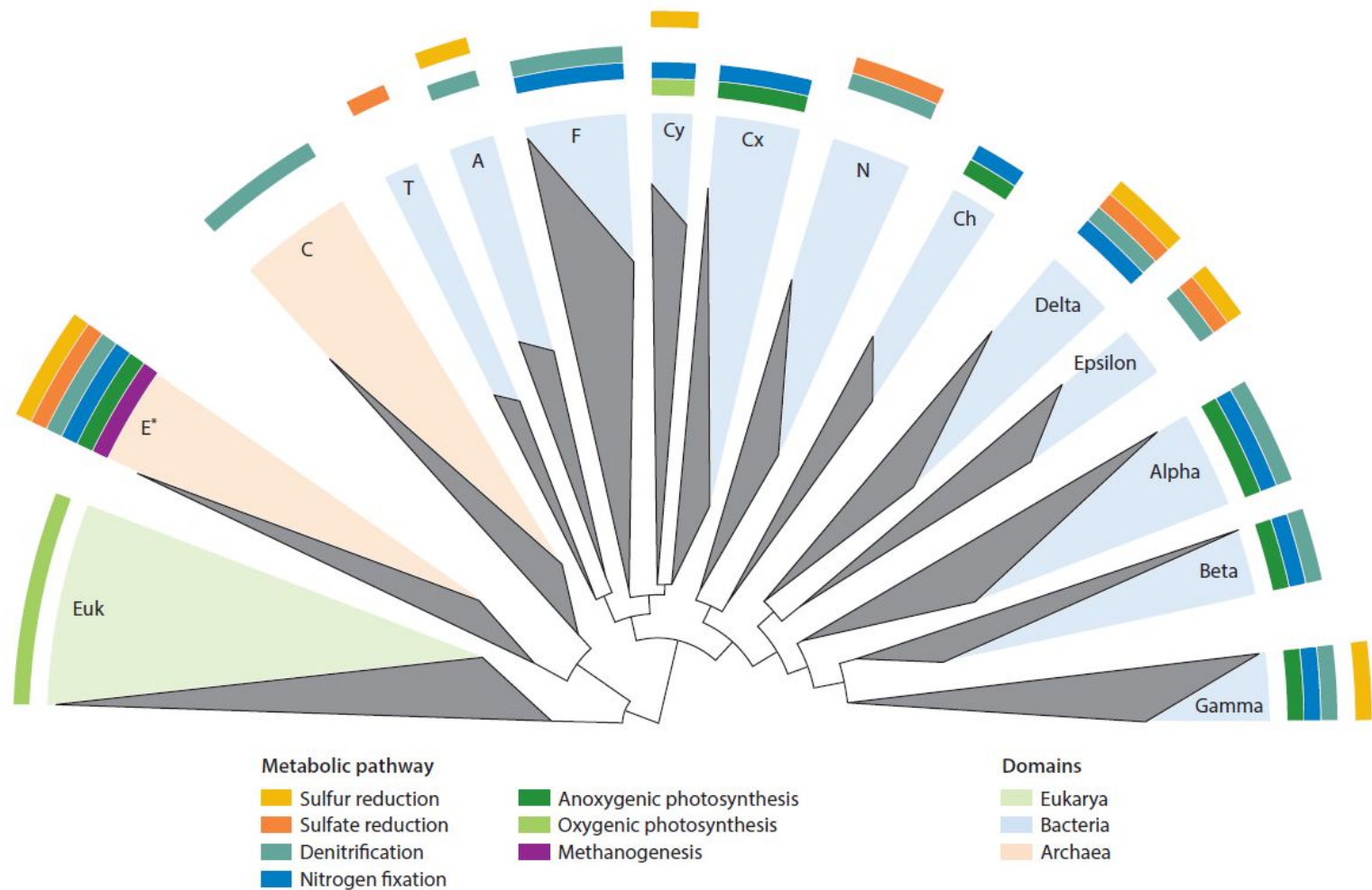
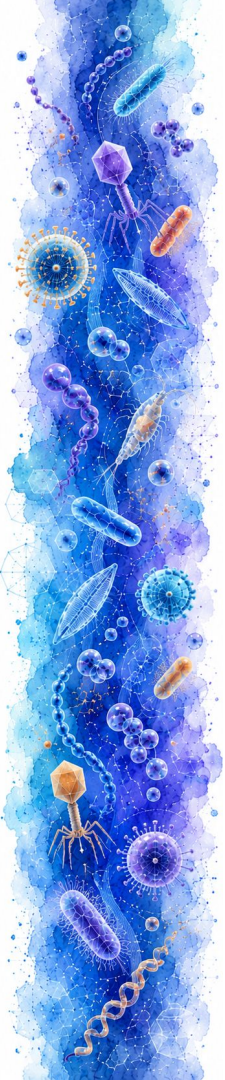
A Periodic Table Of Elements

Group	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Period																		
1	H 1																	He 2
2	Li 3	Be 4											B 5	C 6	N 7	O 8	F 9	Ne 10
3	Na 11	Mg 12											Al 13	Si 14	P 15	S 16	Cl 17	Ar 18
4	K 19	Ca 20	Sc 21	Ti 22	V 23	Cr 24	Mn 25	Fe 26	Co 27	Ni 28	Cu 29	Zn 30	Ga 31	Ge 32	As 33	Se 34	Br 35	Kr 36
5	Rb 37	Sr 38	Y 39	Zr 40	Nb 41	Mo 42	Tc 43	Ru 44	Rh 45	Pd 46	Ag 47	Cd 48	In 49	Sn 50	Sb 51	Te 52	I 53	Xe 54
6	Cs 55	Ba 56		Hf 72	Ta 73	W 74	Re 75	Os 76	Ir 77	Pt 78	Au 79	Hg 80	Tl 81	Pb 82	Bi 83	Po 84	At 85	Rn 86
7	Fr 87	Ra 88		Rf 104	Db 105	Sg 106	Bh 107	Hs 108	Mt 109	Ds 110	Rg 111	Cn 112	Uut 113	Fl 114	Uup 115	Lv 116	Uus 117	Uuo 118
				La 57	Ce 58	Pr 59	Nd 60	Pm 61	Sm 62	Eu 63	Gd 64	Tb 65	Dy 66	Ho 67	Er 68	Tm 69	Yb 70	Lu 71
				Ac 89	Th 90	Pa 91	U 92	Np 93	Pu 94	Am 95	Cm 96	Bk 97	Cf 98	Es 99	Fm 100	Md 101	No 102	Lr 103

A Periodic Table Of Elements

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* Biological categorization taken from Remick KA and Helman JD. (2023) The elements of life: a biocentric tour of the periodic table. *Adv. Microbial Phys.*



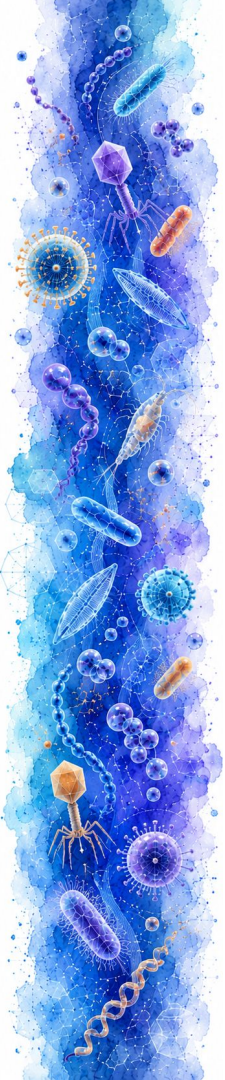


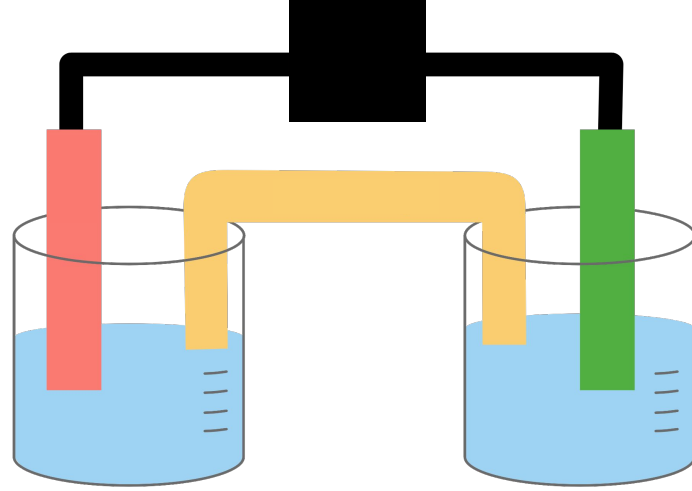
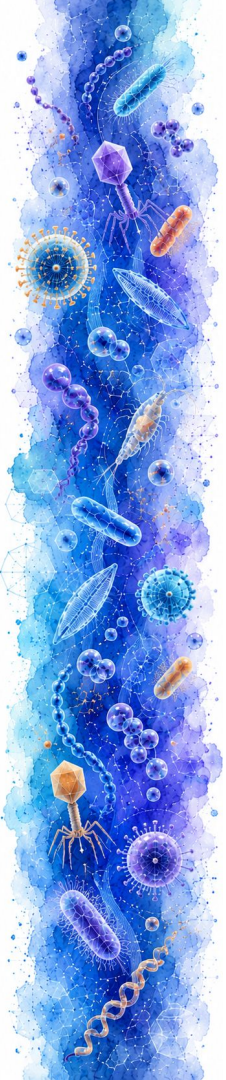
TABLE 3.1 Micronutrients needed by microorganisms^a

<i>I. Trace Elements</i>		<i>II. Growth Factors</i>	
<i>Element</i>	<i>Function</i>	<i>Growth factor</i>	<i>Function</i>
Boron (B)	Autoinducer for quorum sensing in bacteria; also found in some polyketide antibiotics	PABA (<i>p</i> -aminobenzoic acid)	Precursor of folic acid
Cobalt (Co)	Vitamin B ₁₂ ; transcarboxylase (only in propionic acid bacteria)	Folic acid	One-carbon metabolism; methyl transfers
Copper (Cu)	In respiration, cytochrome <i>c</i> oxidase; in photosynthesis, plastocyanin, some superoxide dismutases	Biotin	Fatty acid biosynthesis; some CO ₂ fixation reactions
Iron (Fe) ^b	Cytochromes; catalases; peroxidases; iron–sulfur proteins; oxygenases; all nitrogenases	B ₁₂ (Cobalamin)	One-carbon metabolism; synthesis of deoxyribose
Manganese (Mn)	Activator of many enzymes; component of certain superoxide dismutases and of the water-splitting enzyme in oxygenic phototrophs (photosystem II)	B ₁ (Thiamine)	Decarboxylation reactions
Molybdenum (Mo)	Certain flavin-containing enzymes; some nitrogenases, nitrate reductases, sulfite oxidases, DMSO–TMAO reductases; some formate dehydrogenases	B ₆ (Pyridoxal)	Amino acid/keto acid transformations
Nickel (Ni)	Most hydrogenases; coenzyme F ₄₃₀ of methanogens; carbon monoxide dehydrogenase; urease	Nicotinic acid (Niacin)	Precursor of NAD ⁺
Selenium (Se)	Formate dehydrogenase; some hydrogenases; the amino acid selenocysteine	Riboflavin	Precursor of FMN, FAD
Tungsten (W)	Some formate dehydrogenases; oxotransferases of hyperthermophiles	Pantothenic acid	Precursor of coenzyme A
Vanadium (V)	Vanadium nitrogenase; bromoperoxidase	Lipoic acid	Decarboxylation of pyruvate and α-ketoglutarate
Zinc (Zn)	Carbonic anhydrase; nucleic acid polymerases; many DNA-binding proteins	Vitamin K	Electron transport
		Coenzymes M and B	Methanogenesis ^c
		F ₄₂₀ and F ₄₃₀	Methanogenesis ^c

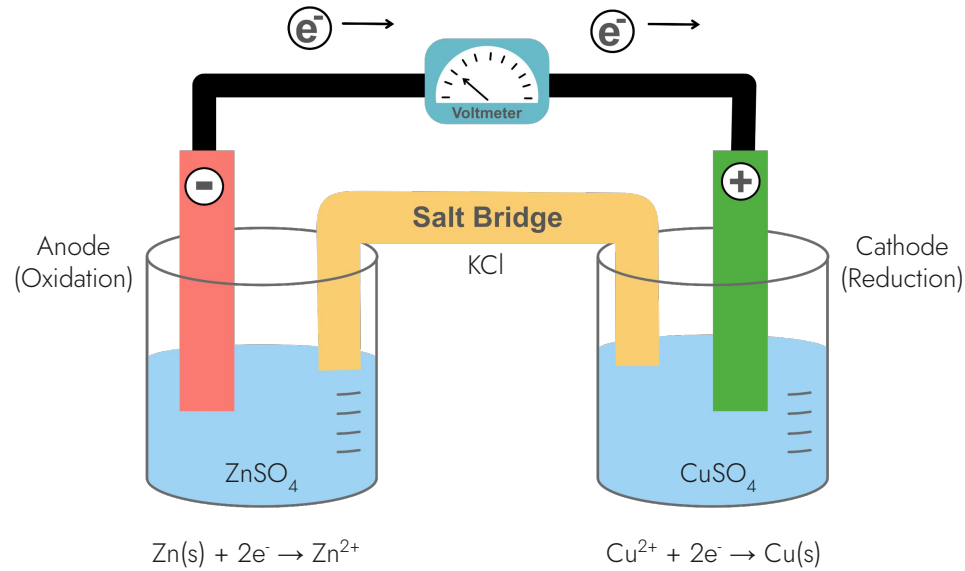
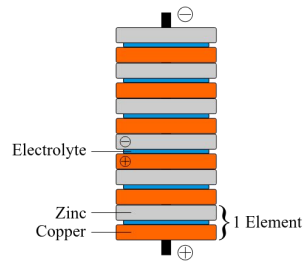
^aNot all trace elements or growth factors are needed by all organisms, and many growth factors are biosynthesized and not required from the environment.

^bIron is typically needed in larger amounts than the other trace metals shown.

^cThe production of methane (CH₄) by methanogens (Archaea).

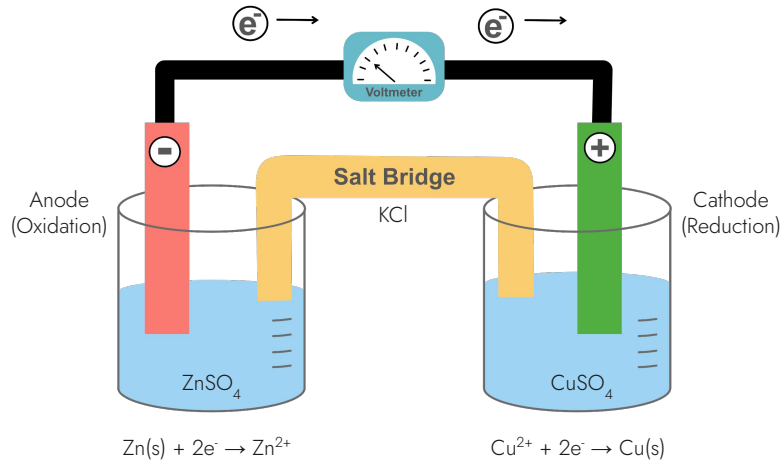


Voltaic Pile

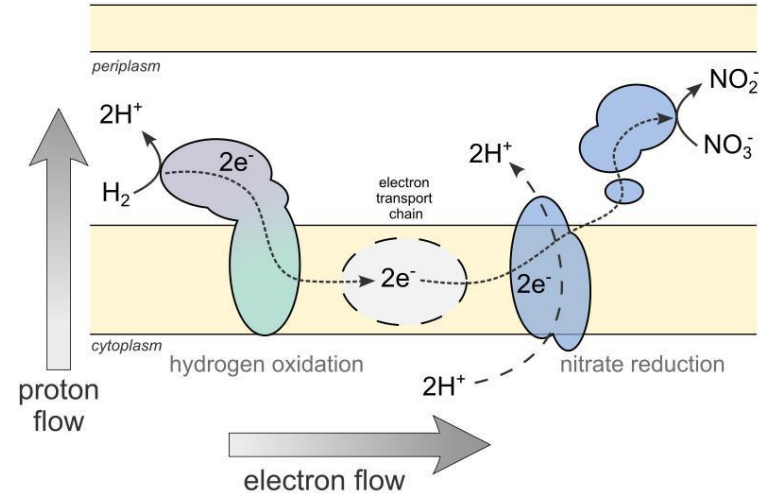


Life Is Electric, Literally

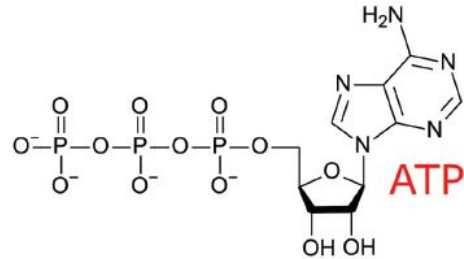
How batteries do



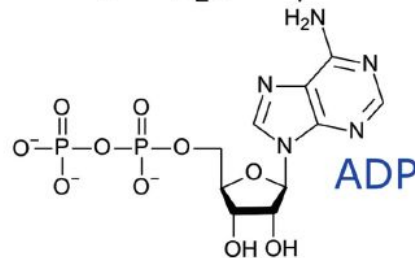
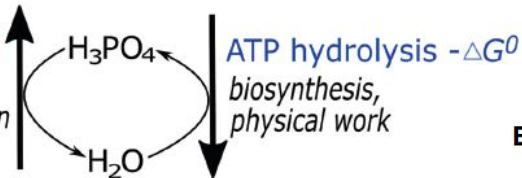
How biology do



Life Building Blocks



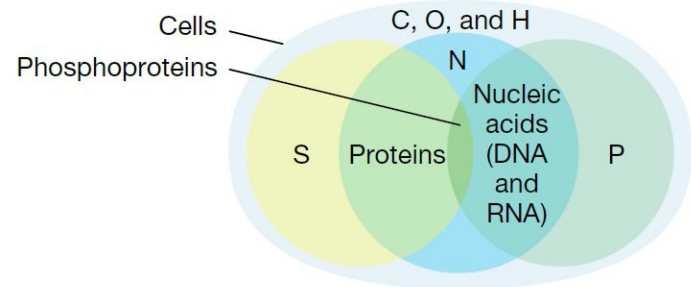
ATP formation $+\Delta G^0$
*glycolysis, TCA cycle,
 oxidative phosphorylation
 photosynthesis*



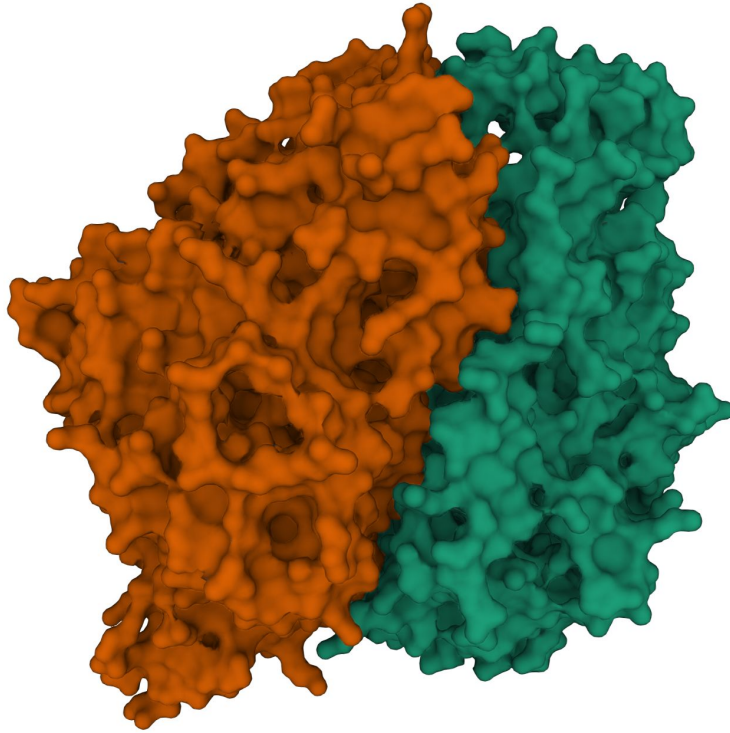
Macromolecular composition of a cell

Macromolecule	Percent of dry weight
Protein	55
Lipid	9.1
Polysaccharide	5.0
Lipopolysaccharide	3.4
DNA	3.1
RNA	20.5

Elemental composition of informational macromolecules

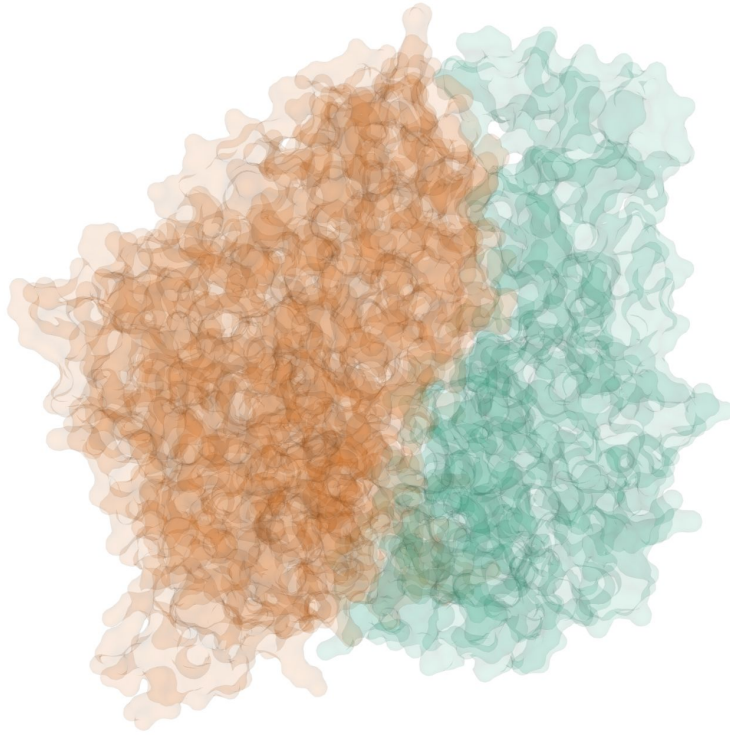


Life Controls Redox Through Metals



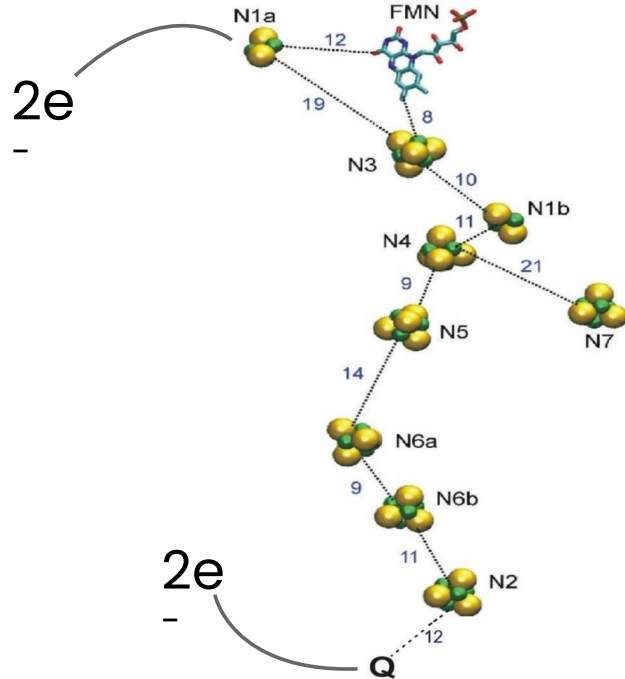
Proteins are just a fancy way to keep the metal centers in the exact position to control redox reactions with minimal energy loss

Life Controls Redox Through Metals



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Life Controls Redox Through Metals



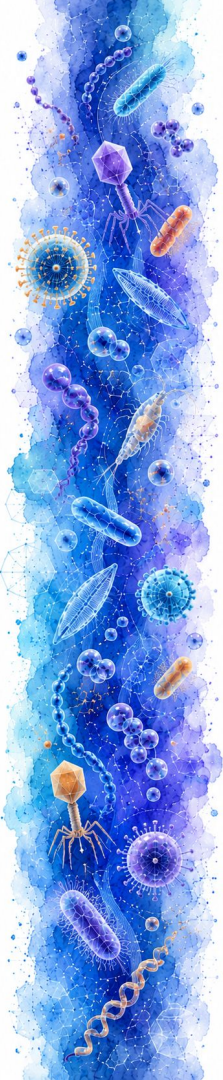
Proteins are just a fancy way to keep the metal centers in the exact position to control redox reactions with minimal energy loss



Still Talking About Science Here?

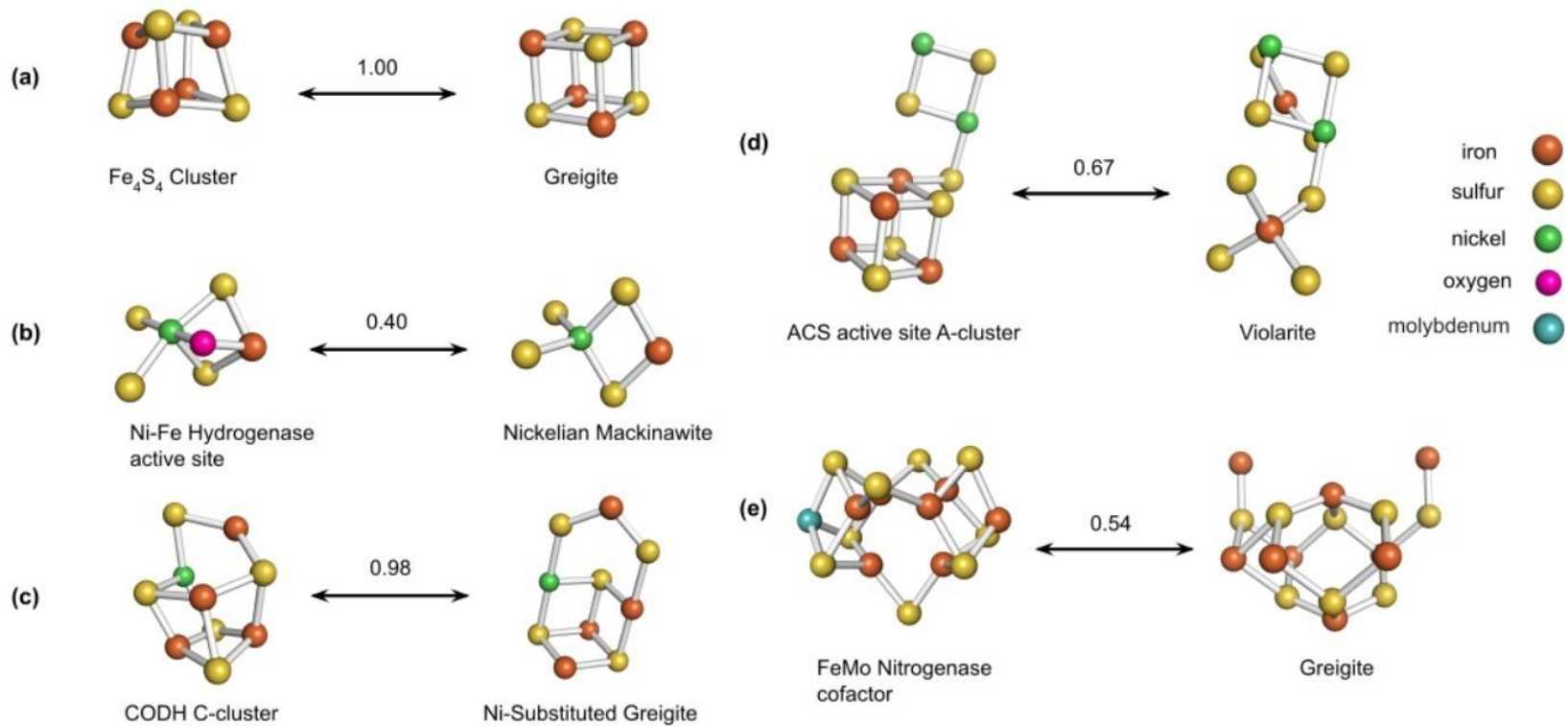
Learning from the Marvel Universe

Infinite stones empowering cosmic energy must be placed the Infinity Gauntlet that acts as a **protective conduit**, amplifier, and **control interface** that allows a mortal or alien body to **channel the** raw, lethal **energy** of all six Infinity Stones at once



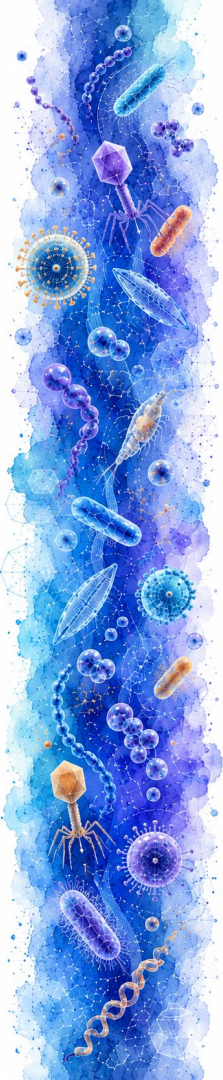
BIOLOGY

GEOLOGY

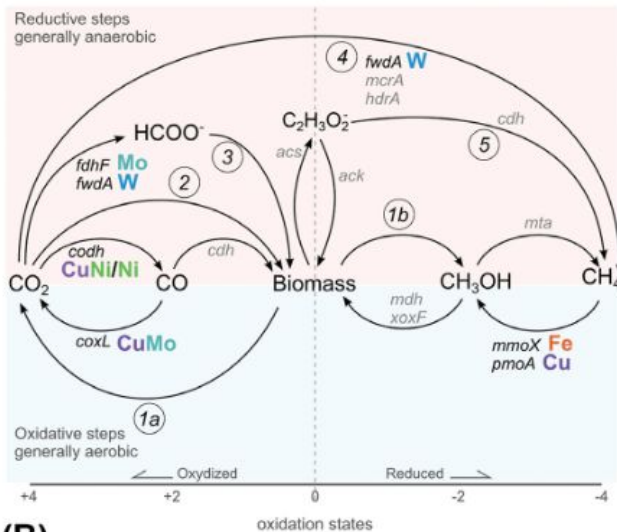


BIOLOGY

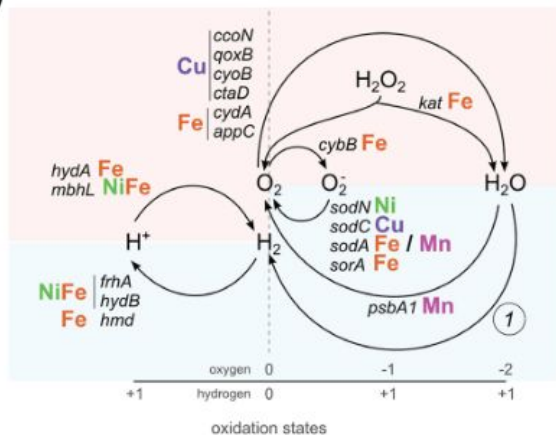
GEOLOGY



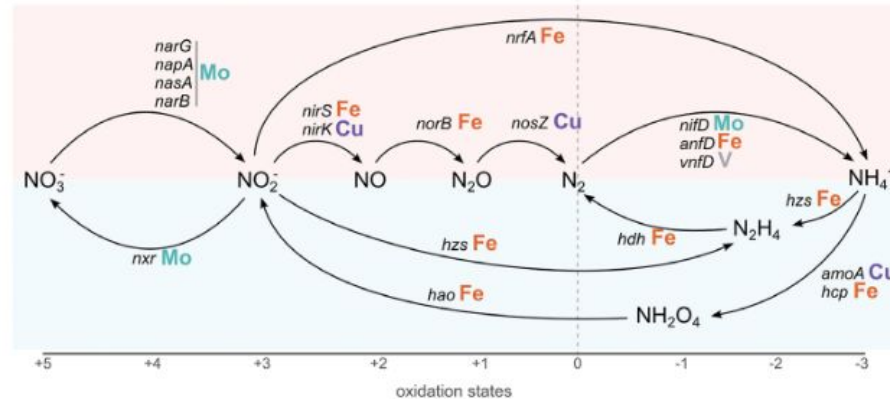
(A)



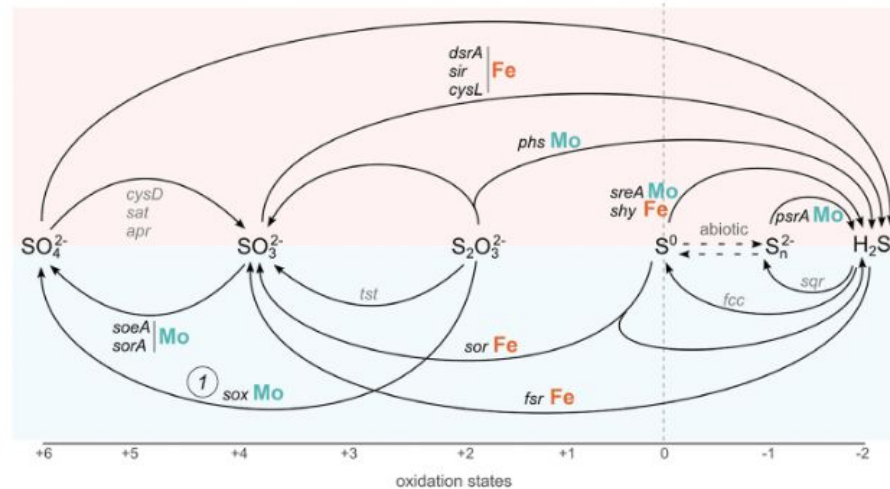
(B)

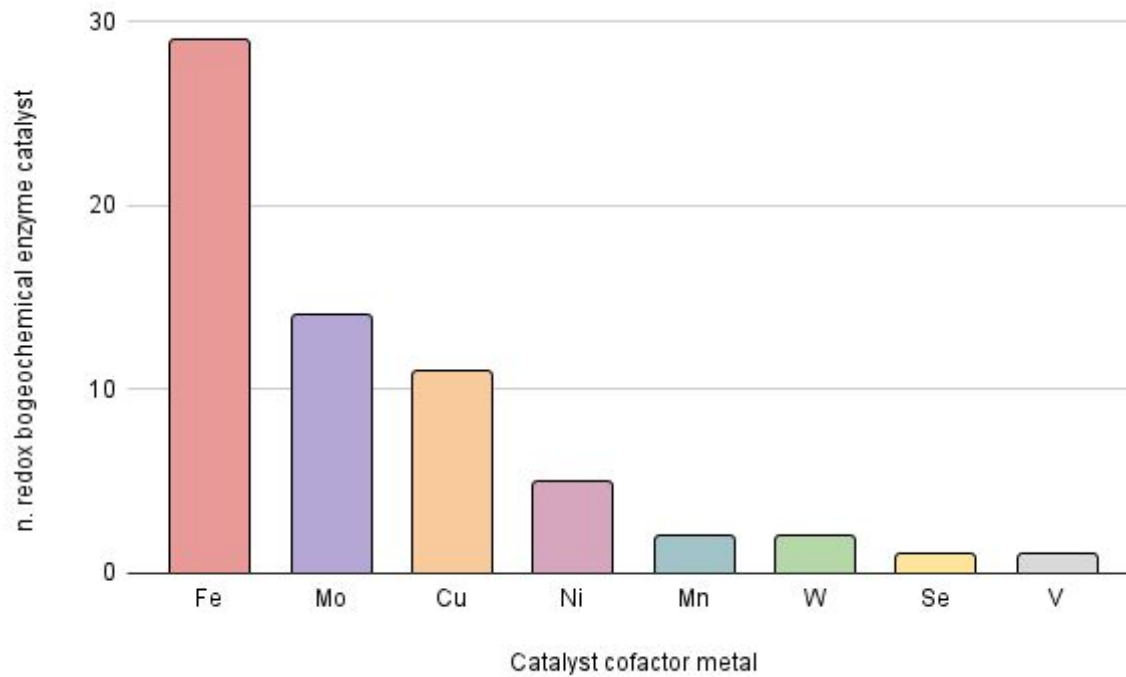
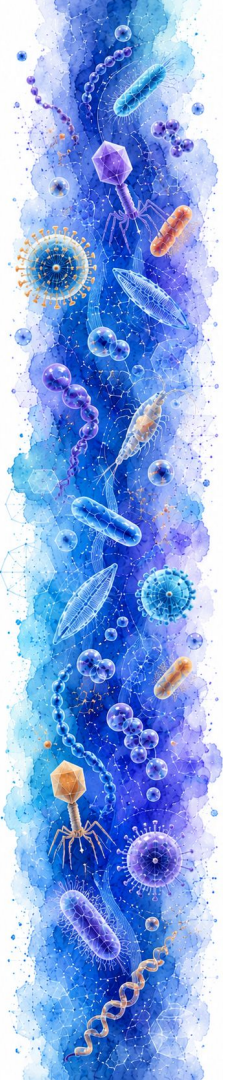


(C)



(D)



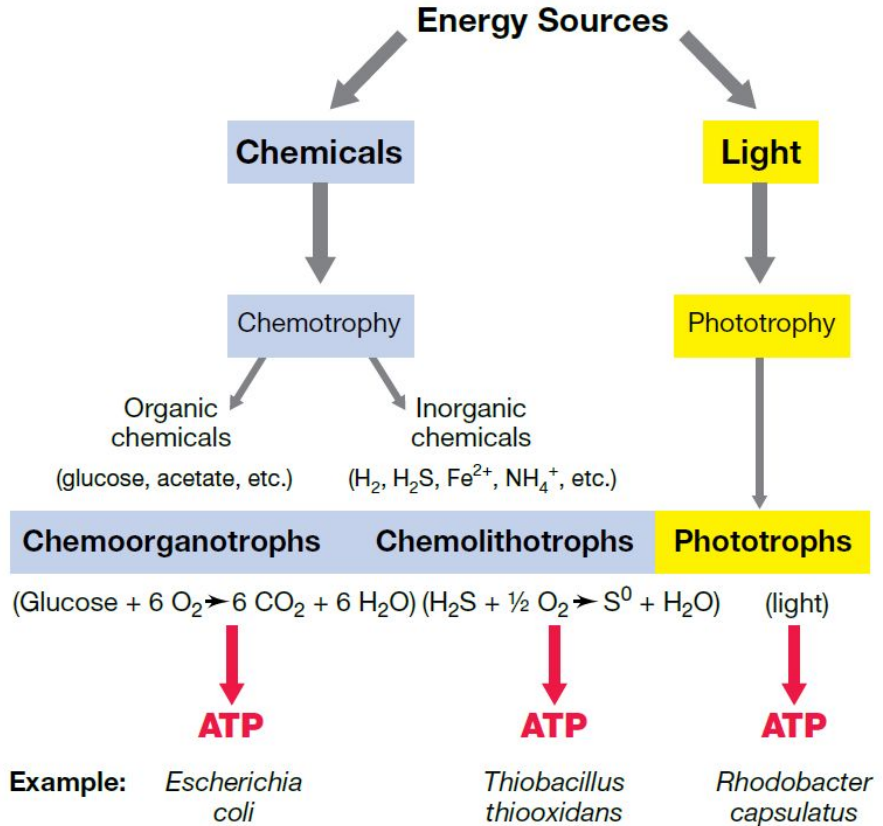


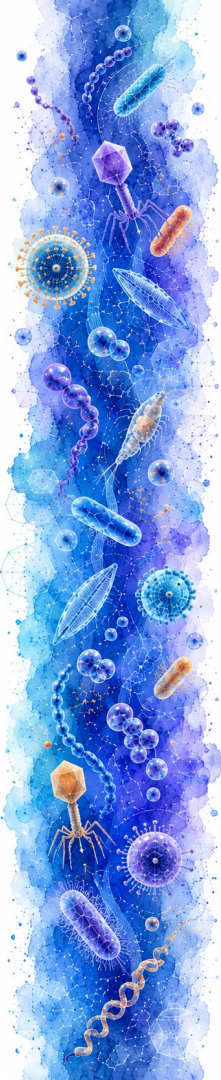
Life Is Electric

Living organisms require energy to sustain life. Life uses **energy** in two forms, **chemical** and **light**, and is needed to fuel various **processes** like **growth**, **repair**, **movement**, and **metabolism**. This free energy is not just passively consumed. It is actively transferred from one chemical reaction to another.

A major way cells achieve energy transfer is through **redox reactions**.

In these reactions, electrons are transferred between **donors** and **acceptors**. This electron flows generate energy used to pump protons across the membrane, creating an electrochemical gradient that activates the synthesis of ATP as energy storage.

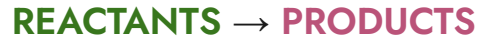




Principles of Bioenergetics

In microbiology, free energy (**G**) is defined as the **energy available to do work**, and energy transformations are measured in kilojoules (kJ), a unit of heat energy.

Free energy of formation (G_f^0), or Gibbs energy, refers to the **energy yielded or required** for the **formation of a given molecule** from its constituent elements.



The **change in free energy** during a reaction that occur under **standard conditions** (pH 7, 25 °C, 1 atm, [reactants] = 1 M) is expressed as ΔG^0

Negative values ΔG^0 = energy release = exergonic reaction

Positive values ΔG^0 = energy requirement = endergonic reaction

Principles of Bioenergetics

TABLE 3.2 Free energy of formation (G_f^0 , kJ/mol) for some common substances^a

<i>Sugars</i>	<i>Organic and fatty acids</i>	<i>Amino acids and alcohols</i>	<i>Gases and inorganic compounds</i>
Fructose (−951.4)	Acetate (−369.4)	Alanine (−371.5)	O ₂ , N ₂ , H ₂ , S ⁰ , Fe ⁰ (0)
Glucose (−917.2)	Benzoate (−245.6)	Aspartate (−700.4)	CH ₄ (−50.8)
Lactose (−1515.2)	Butyrate (−352.6)	<i>n</i> -Butanol (−171.8)	CO ₂ (−394.4); CO (−137.4)
Ribose (−369.4)	Caproate (−335.9)	Ethanol (−181.7)	H ₂ O (−237.2); H ⁺ (−39.8); OH [−] (−198.7)
Sucrose (−757.3)	Citrate (−1168.3)	Glutamate (−699.6)	N ₂ O (+104.2); NO (+86.6)
	Formate (−351.1)	Glutamine (−529.7)	NO ₂ [−] (−37.2); NO ₃ [−] (−111.3)
	Fumarate (−604.2)	Glycerol (−488.5)	NH ₃ (−26.57); NH ₄ ⁺ (−79.4)
	Glyoxylate (−468.6)	Mannitol (−942.6)	H ₂ S (−27.87); HS [−] (+12.1)
	Ketoglutarate (−797.5)	Methanol (−175.4)	SO ₄ ^{2−} (−744.6); S ₂ O ₃ ^{2−} (−513.4)
	Lactate (−517.8)	<i>n</i> -Propanol (−175.8)	Fe ²⁺ (−78.8); Fe ³⁺ (−4.6); FeS (−100.4)
	Malate (−845.1)		
	Propionate (−361.1)		
	Pyruvate (−474.6)		
	Succinate (−690.2)		
	Valerate (−344.3)		

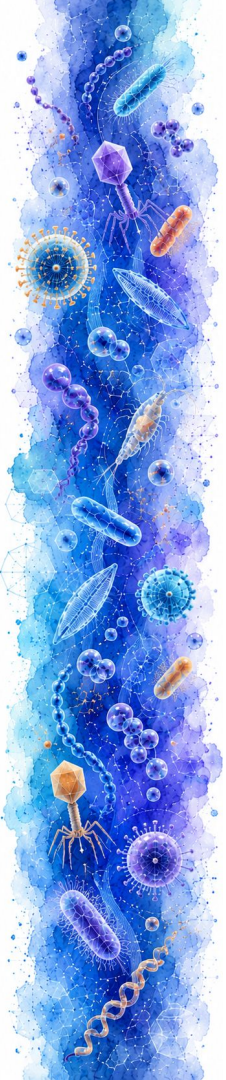
^aValues for free energy of formation taken from Speight, J. 2005. *Lange's Handbook of Chemistry*, 16th edition, and Thauer, R.K., K. Jungermann, and H. Decker. 1977. Energy conservation in anaerobic chemotrophic bacteria. *Bacteriol. Rev.* 41: 100–180.

Principles of Bioenergetics

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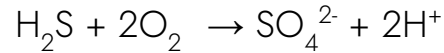
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Principles of Bioenergetics

REACTANTS PRODUCTS



$$G_f^0 \text{H}_2\text{S} = -27.87 \text{ kJ/mol}$$

$$G_f^0 \text{O}_2 = 0 \text{ kJ/mol}$$

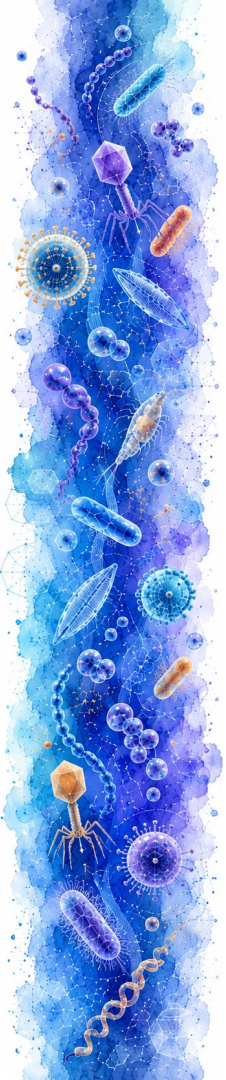
$$G_f^0 \text{SO}_4^{2-} = -744.6 \text{ kJ/mol}$$

$$G_f^0 \text{H}^+ = -39.83 \text{ kJ/mol}$$

$$\Delta G_0' = (-744.6) + 2(-39.83) - (-27.87) + (0)$$

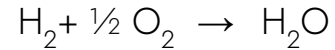
$$\Delta G_0' = -796.39 \text{ kJ/mol}$$

EXERGONIC REACTION → SPONTANEOUS



Principles of Bioenergetics

REACTANTS **PRODUCTS**



$$G_f^0 \text{H}_2 = 0 \text{ kJ/mol}$$

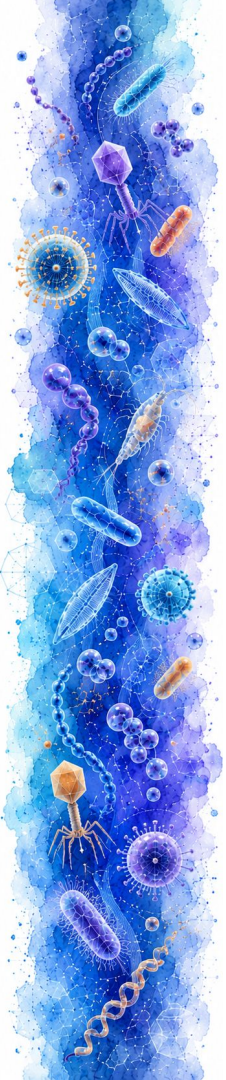
$$G_f^0 \text{O}_2 = 0 \text{ kJ/mol}$$

$$G_f^0 \text{H}_2\text{O} = -744.6 \text{ kJ/mol}$$

$$\Delta G^0' = -237.2 - 0 + \frac{1}{2} (0)$$

$$\Delta G^0' = -237.2 \text{ kJ/mol}$$

EXERGONIC REACTION → NEED A CATALYST



Thermodynamics VS Kinetics

Some exergonic reactions, despite having a negative Gibbs free energy and being thermodynamically favorable, are not spontaneous due to some kinetic factors.

.High activation energy

Even though the reaction releases energy overall, it may have a large activation **energy barrier** that **must be overcome** for the reaction to proceed. Without sufficient energy to overcome this activation barrier (e.g., through heat or the presence of a catalyst), the reaction will not proceed spontaneously.

.Reaction rate

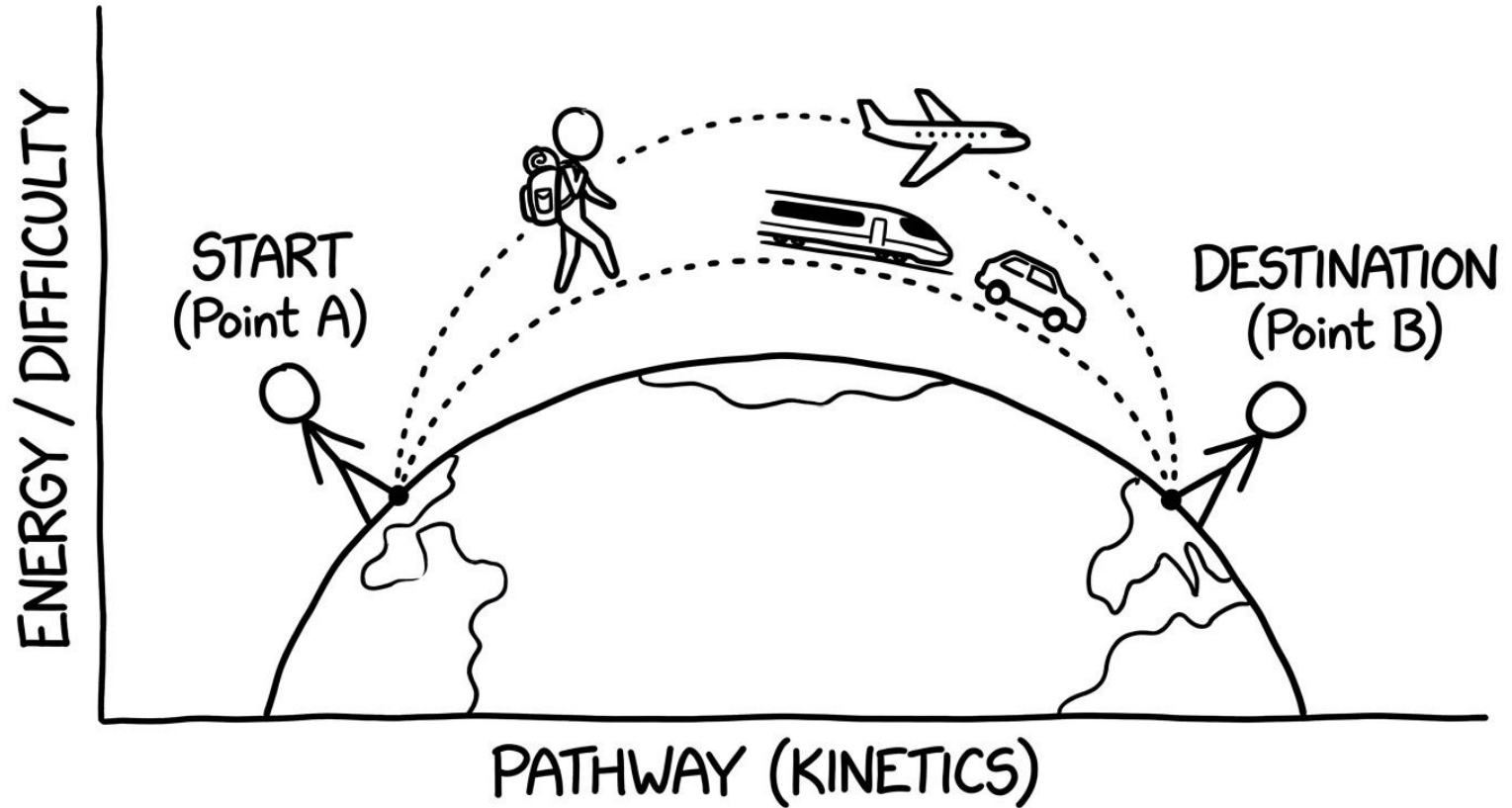
The **spontaneity** of a reaction based on free energy **does not account for how quickly the reaction occurs**. Some reactions are so slow at room temperature that they effectively don't happen without a catalyst, even if they are thermodynamically favorable.

.Reaction conditions

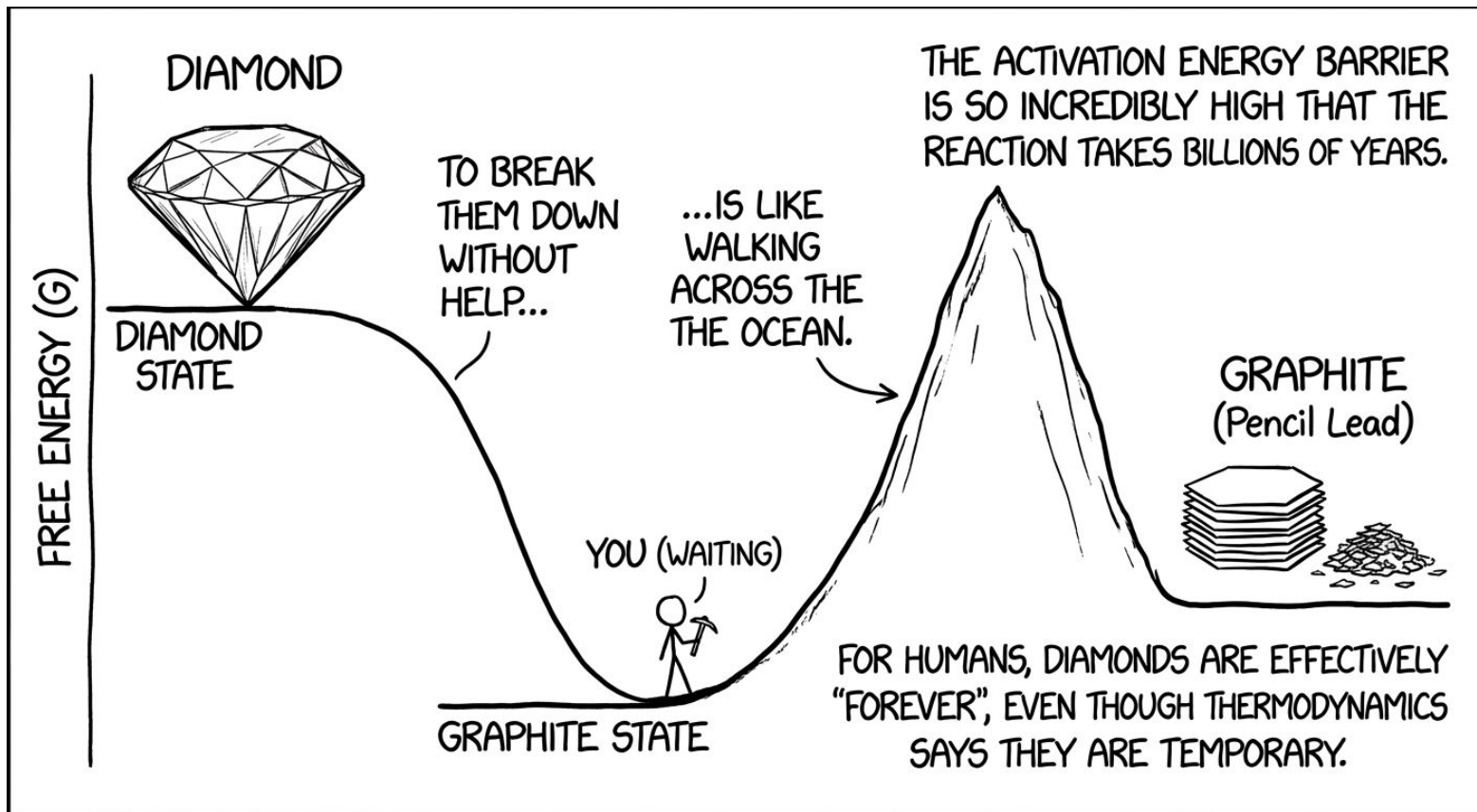
The actual **spontaneity** of a reaction can depend on environmental conditions such as **temperature**, **pressure**, and **concentration of reactants**. A reaction may be exergonic under certain conditions but not under others.

Thus, the kinetics of a reaction are as important as the thermodynamics in determining whether a reaction will proceed spontaneously.

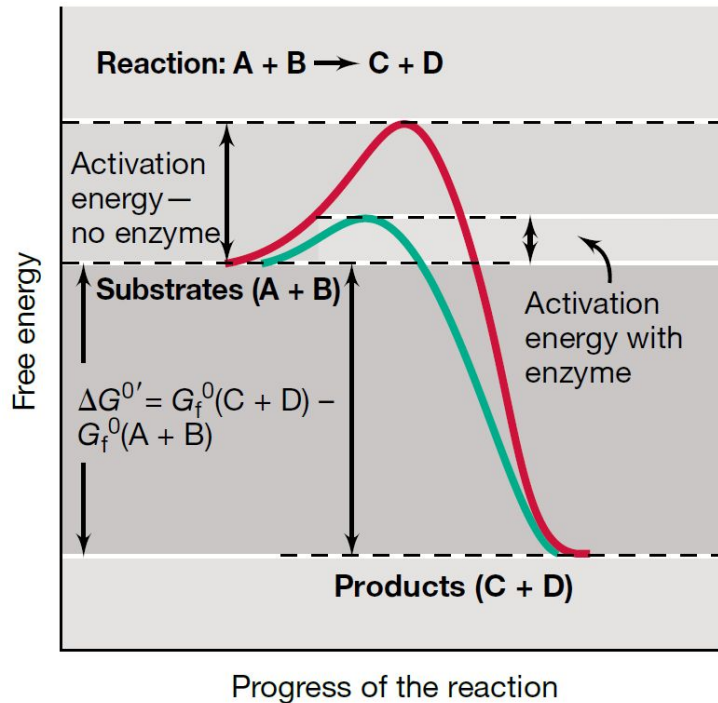
Thermodynamics VS Kinetics

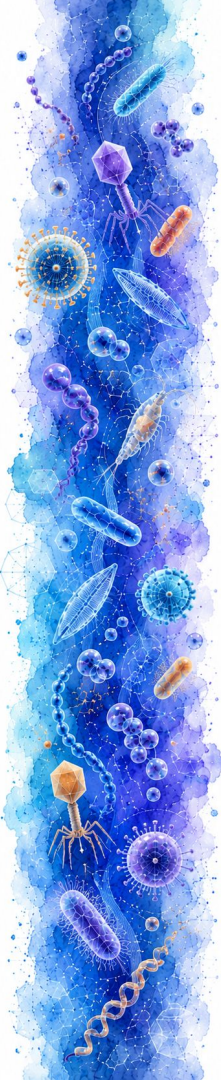


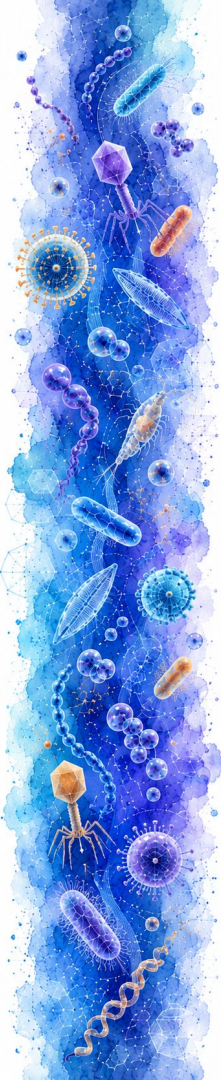
Thermodynamics VS Kinetics



Why The Enzymes?

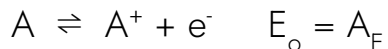


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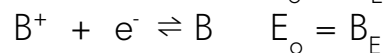


Thermodynamics VS Kinetics

An oxidation–reduction or **redox reaction** is a reaction that involves the **transfer of electrons** between chemical species, where electrons are passed **from an electron donor to an electron acceptor**. Every **redox reaction** can be divided into **2 half-cell reactions**, in which one compound **loses** electrons (**oxidation**) and another compound **gains** electrons (**reduction**).



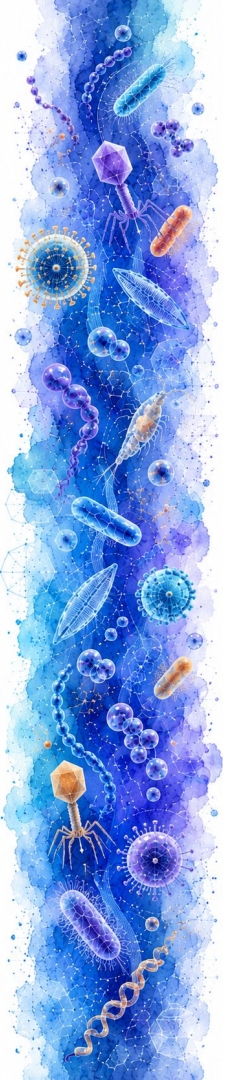
Oxidation half reaction: A is oxidized to A^+



Reduction half reaction: B^+ is reduced to B

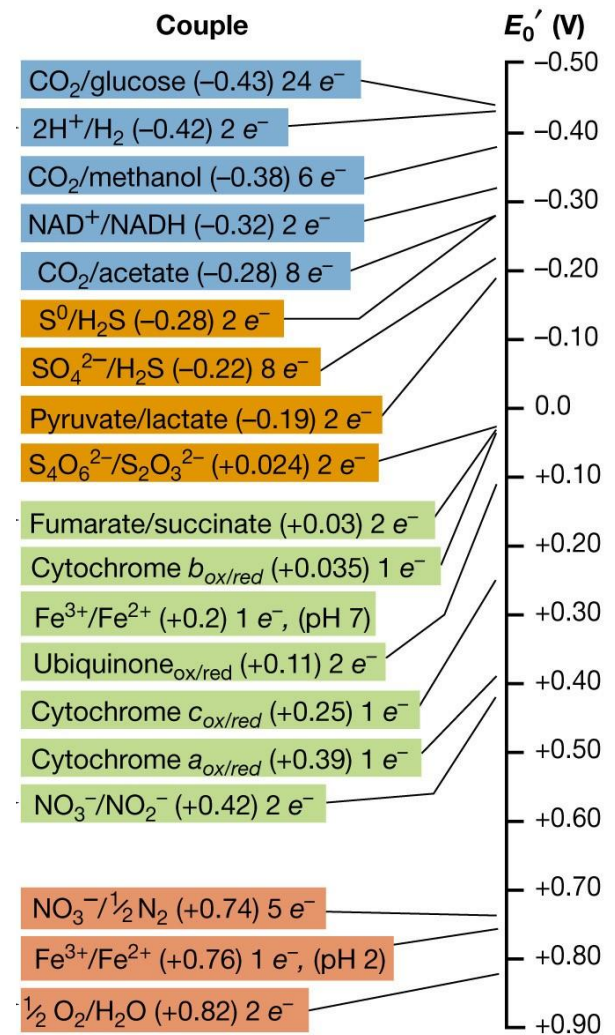
Substances **differ in their tendency** to donate or accept electrons.

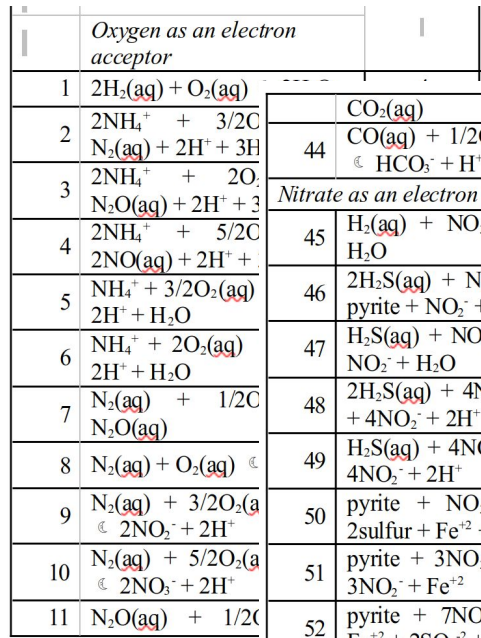
This tendency is represented by their **reduction potential (E_o')**, which is the potential, expressed in volt (V), to gain an electron under standard conditions. The more positive the reduction potential, the greater the affinity for electrons and tendency to be reduced.



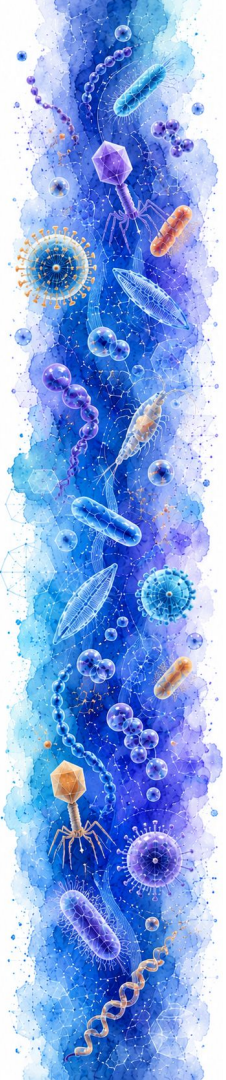
The Redox Tower

In a redox tower, the substances are listed according to their reduction potential (E°), which measures their tendency to gain or lose electrons in redox reactions. Each substance in the tower exists as a **redox couple**, represented as the oxidized/reduced form (e.g., $\text{SO}_4^{2-}/\text{H}_2\text{S}$). The **top** of the tower contains the **strongest electron donors** (those that are in their reduced form and are easily oxidized). These molecules have **negative reduction potentials**, meaning they have a **strong tendency to give electrons**. At the bottom of the tower are the strongest electron acceptors (those that are in their oxidized form and are easily reduced). These have **positive reduction potentials**, meaning they have a **strong tendency to accept electrons**. The **larger** the **difference** in reduction potential between the donor and acceptor, the **more energy** is released when electrons are transferred from the donor to the acceptor. For example, when electrons are transferred from a strong donor (like glucose) to a strong acceptor at the bottom (like O_2), a large amount of free energy is released, which can be harnessed by cells to do work, such as generating ATP.



[illegible]

Using only CHNOS and Fe
Everett Shock group wrote 1,460
thermodynamic valid redox reactions
potentially used by microbes!



Recap

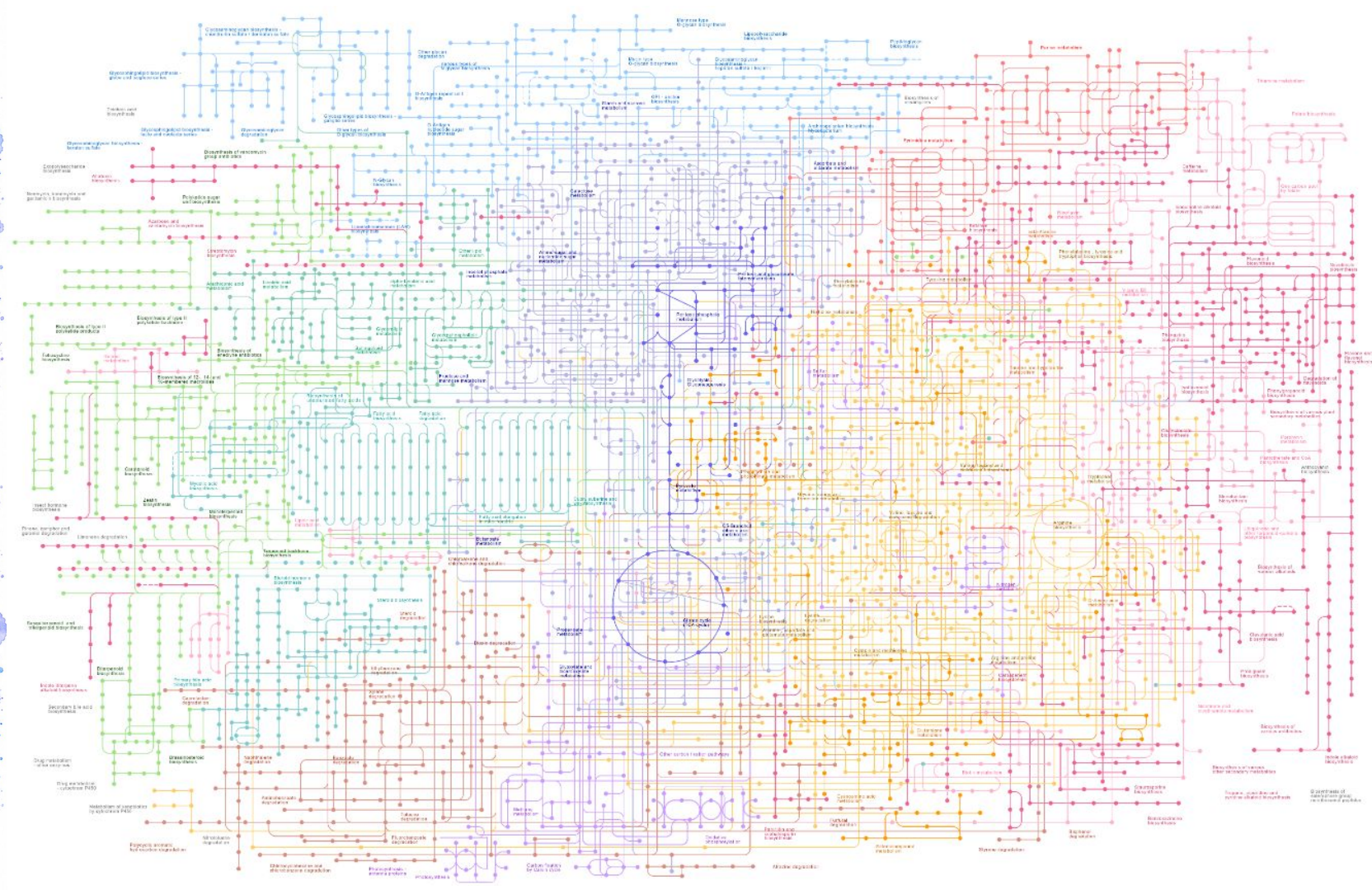
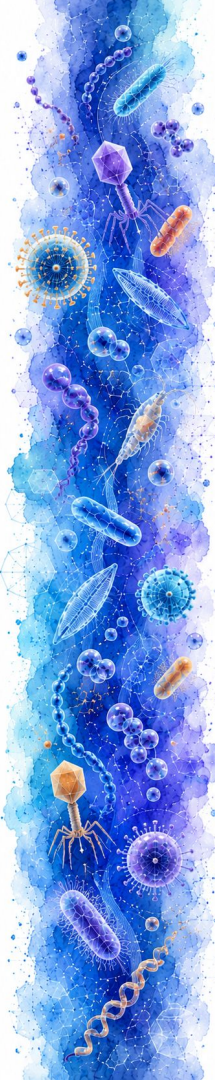
- .Free energy of formation (G_f^0) refers to the the **energy yielded or required** for the formation of a given molecule from its constituent elements
- .The **change in free energy during a reaction**, ΔG^0 , indicates if the reaction occur with a release of energy (**exergonic**) or require energy (**endergonic**)
- .The **more negative** ΔG^0 , the **higher energy release**
- .**Reduction potential** (E^0) is the tendency of a substance to **gain** an electron
- .The **more positive** the E^0 , the **greater** the **affinity** for electrons and tendency to be reduced
- .The **difference** in E^0 between the **donor** and **acceptor redox couples** is quantified as ΔE^0
- .In a redox, the **greater** is ΔE^0 between the **two redox couples** and the **greater** is the **amount of energy released**
- .The relation between the voltage and the energy of a reaction is given by the Nernst equation: $\Delta G^0 = -nF\Delta E^0$
(n = number of moles of electrons transferred in the redox reaction; F = Faraday's constant:96,485 C/mol)

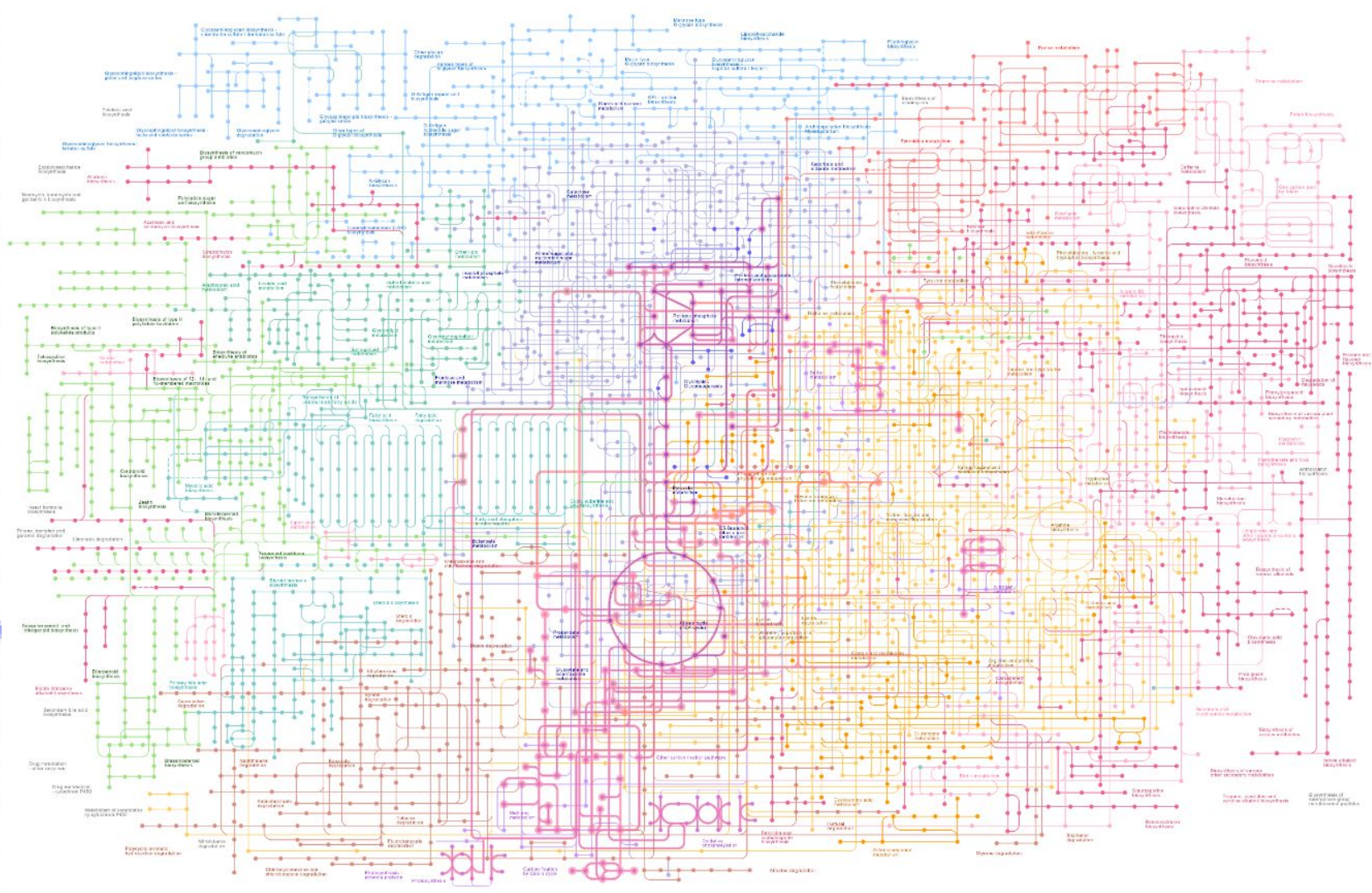
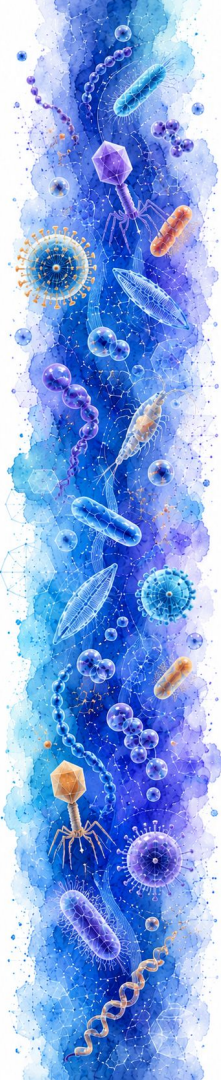
TO BE CONTINUED...



10:00

MakeAGIF.com

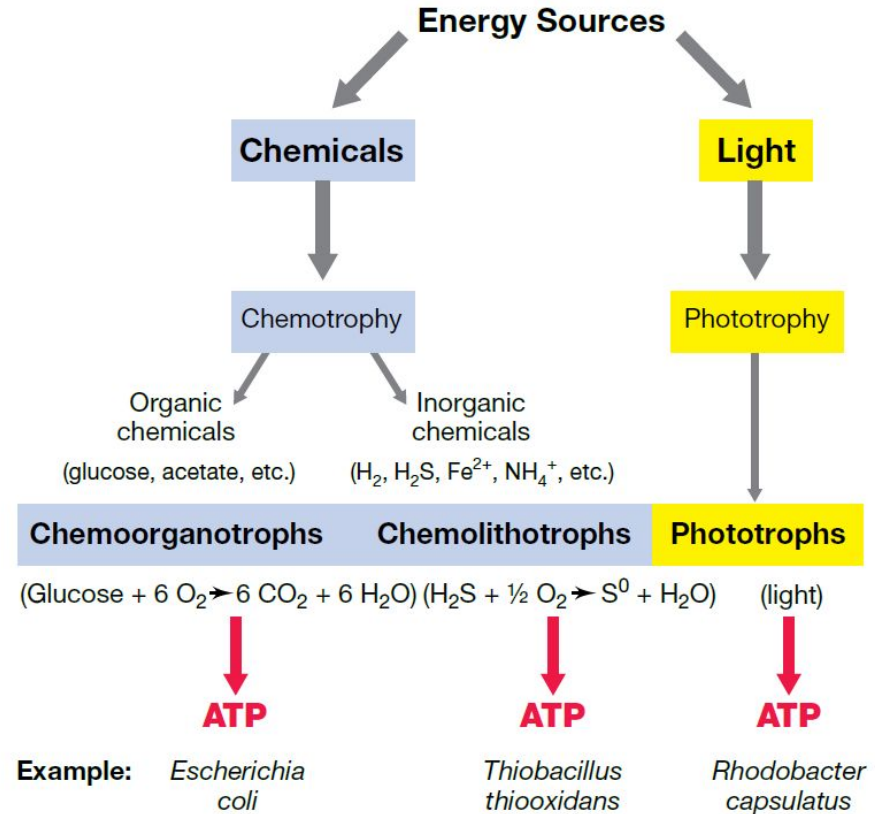




Basic Metabolism

All types of metabolism require these three basic elements:

- .An electron donor (the source of reducing power used to carry out redox reactions)
- .A carbon source used as a donor of carbon for biosynthetic purposes
- .An electron acceptor, an oxidizing agent used to dispose of excess reducing equivalents

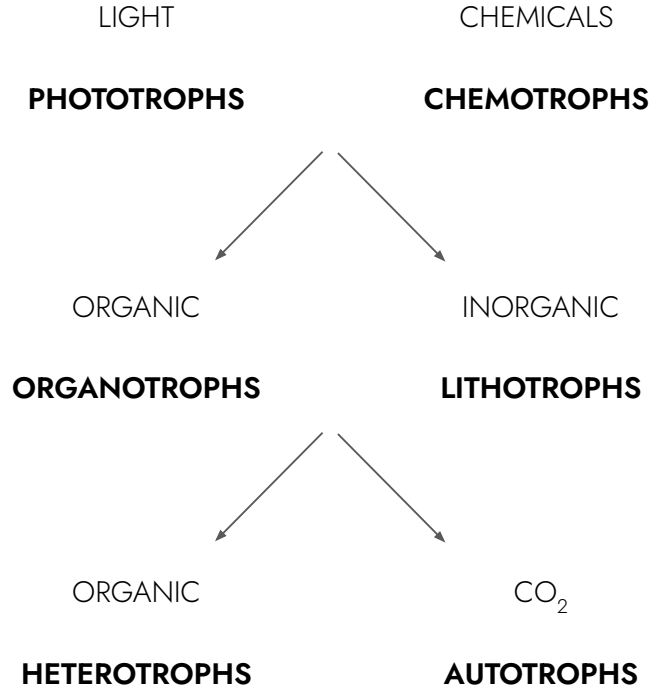


Metabolic Lifestyles

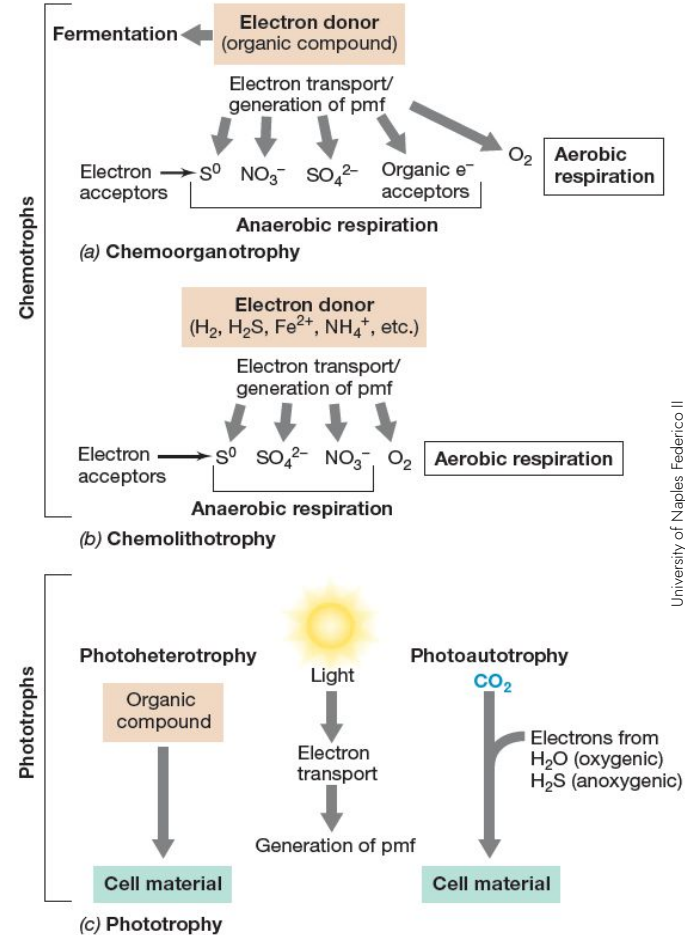
ENERGY
source

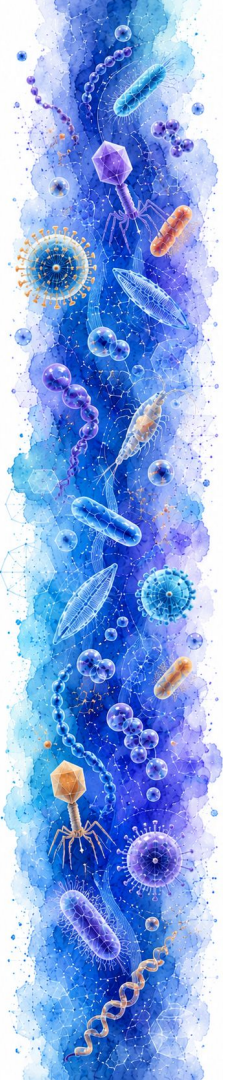
ELECTRONS
donors

CARBON
source



Special cases:
Methylotrophy, Methanogenesis, Syntrophy



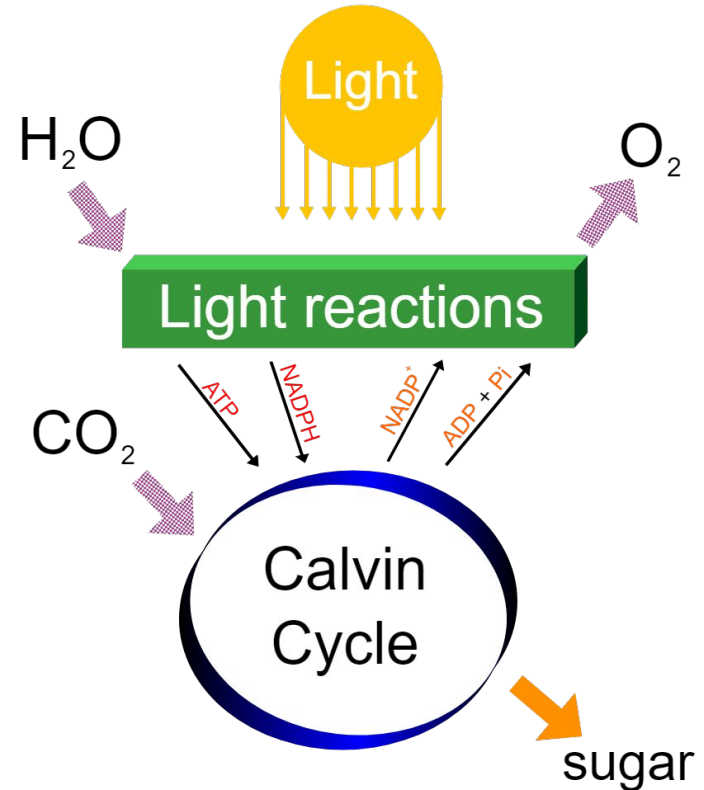


Energy source	Carbon source	Hydrogen or electron source	Representative examples
Photolithoautotrophy			
Light	CO ₂	Inorganic	Cyanobacteria Purple sulfur bacteria Phototrophic protists
Photoorganoheterotrophy			
Light	Organic compounds	Organic compounds or H ₂	Purple non-sulfur bacteria Aerobic anoxygenic bacteria Proteorhodopsin-containing bacteria and archaea ^a
Chemolithoautotrophy			
Inorganic	CO ₂	Inorganic	Sulfur-oxidizing bacteria Hydrogen bacteria Methanogens Nitrifying bacteria and archaea
Chemoorganoheterotrophy			
Organic compounds	Organic compounds	Organic compounds	Wide range of bacteria and archaea Fungi Phagotrophic protists
Mixotrophy (combination of lithoautotrophy and organoheterotrophy)			
Inorganic	Organic compounds	Inorganic	Some sulfur-oxidizing bacteria, e.g. <i>Beggiatoa</i>
Mixotrophy (combination of photoautotrophy and organoheterotrophy)			
Light + organic compounds	CO ₂ + organic compounds	Inorganic or organic	Phagotrophic photosynthetic protists (some flagellates and dinoflagellates)

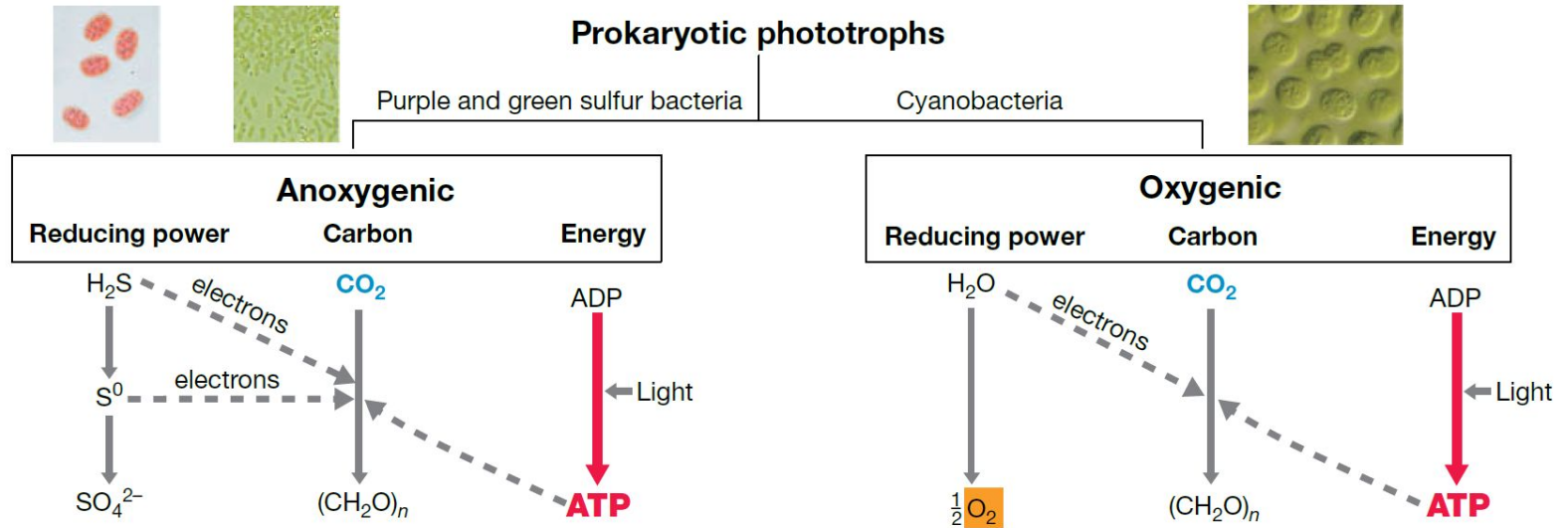
Phototrophs

Phototrophs are organisms that carry out photosynthesis, the conversion of light energy to chemical energy. They are generally **photolithoautotrophs**, in which the **energy** is obtained through **light-driven reactions** involving the **photo-oxidation** of an **inorganic electron donor** (H_2O). This energy is then used in the carbon fixation process, in which an **inorganic carbon source** (CO_2) is reduced to organic compounds through the **Calvin cycle**.

Photosynthesis is the **most important contributor to primary productivity** accounting for the major roles played by **cyanobacteria**, **diatoms**, and **dinoflagellates** and **other protists** in ocean processes.



Photolithoautotrophs

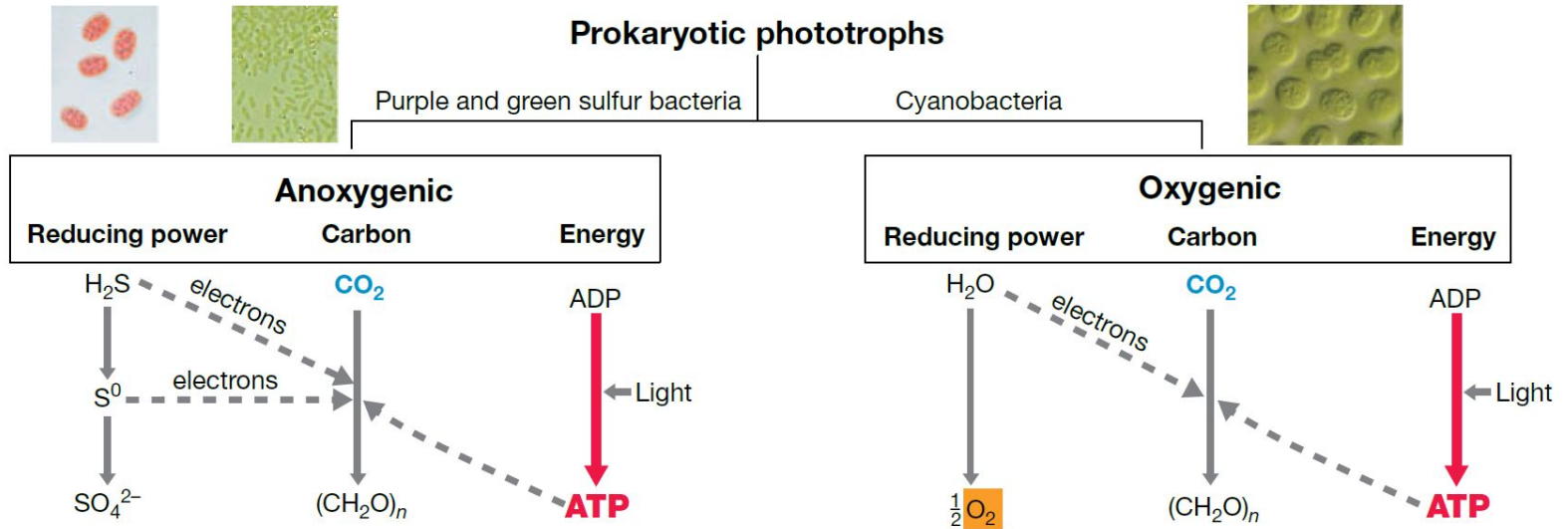


Prokaryotic photosynthesis occurs in photosynthetic membranes called **chromatophores** in purple bacteria, **thylakoids** in cyanobacteria and **chlorosome** in green sulfur bacteria.

Anoxygenic phototrophs use electron donors like H_2S , H_2 , Fe^{2+} , Formate, NO_2^- , bacteriochlorophyll-g



Photolithoautotrophs

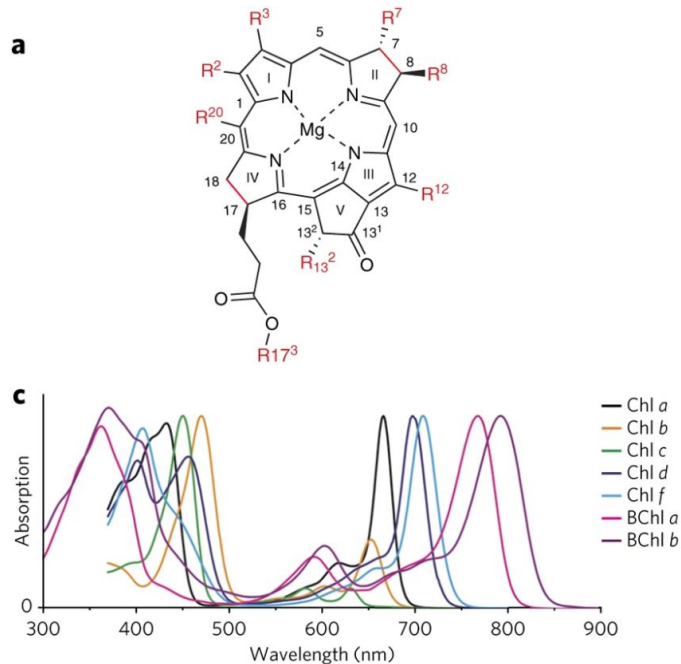


Prokaryotic photosynthesis occurs in photosynthetic membranes called **chromatophores** in purple bacteria, **thylakoids** in cyanobacteria and **chlorosome** in green sulfur bacteria.

Oxygenic photosynthesis is the conversion of sunlight into chemical energy, in which H_2O is used as electron donor and O_2 is the waste product of water oxidation



Light-Sensitive Pigments

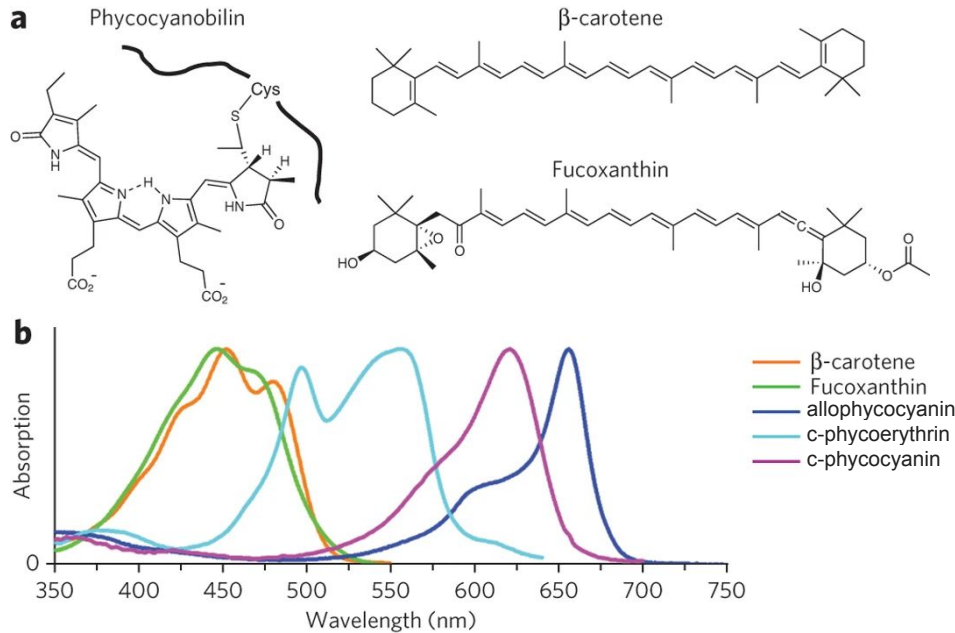


b

Pigment	R2	R3	R7	R8	7-8 bond	R12	R13 ²	17-18 bond	R20	R17 ³
Chl <i>a</i>	CH ₃	CH=CH ₂	CH ₃	C ₂ H ₅	=	CH ₃	COOCH ₃	-	H	Phytol
Chl <i>b</i>	CH ₃	CH=CH ₂	CHO	C ₂ H ₅	=	CH ₃	COOCH ₃	-	H	Phytol
Chl <i>c</i> 1	CH ₃	CH=CH ₂	CH ₃	C ₂ H ₅	=	CH ₃	COOCH ₃	=	H	See note
Chl <i>c</i> 2	CH ₃	CH=CH ₂	CH ₃	CH=CH ₂	=	CH ₃	COOCH ₃	=	H	See note
Chl <i>c</i> 3	CH ₃	CH=CH ₂	COOCH ₃	CH=CH ₂	=	CH ₃	COOCH ₃	=	H	See note
Chl <i>d</i>	CH ₃	CHO	CH ₃	C ₂ H ₅	=	CH ₃	COOCH ₃	-	H	Phytol
Chl <i>f</i>	CHO	C ₂ H ₅	CH ₃	C ₂ H ₅	=	CH ₃	COOCH ₃	-	H	Phytol
BChl <i>a</i>	CH ₃	COCH ₃	CH ₃	C ₂ H ₅	-	CH ₃	COOCH ₃	-	H	Phytol
BChl <i>b</i>	CH ₃	COCH ₃	CH ₃	=CH-CH ₃	-	CH ₃	COOCH ₃	-	H	Phytol
BChl <i>c</i>	CH ₃	CHOH-CH ₃	CH ₃	Various	=	Various	H	-	CH ₃	Various
BChl <i>d</i>	CH ₃	CHOH-CH ₃	CH ₃	Various	=	Various	H	-	H	Various
BChl <i>e</i>	CH ₃	CHOH-CH ₃	CHO	Various	=	C ₂ H ₅	H	-	CH ₃	Various

The **main pigments** used in natural light harvesting are substituted tetrapyrroles, including **chlorophylls** and **bacteriochlorophylls**. The absorption **properties of these pigments vary** owing to the extent of the conjugation and the number and nature of the substitutions, **creating a 'palette' that covers the visible and infrared regions of the solar spectrum.**

Light-Sensitive Pigments



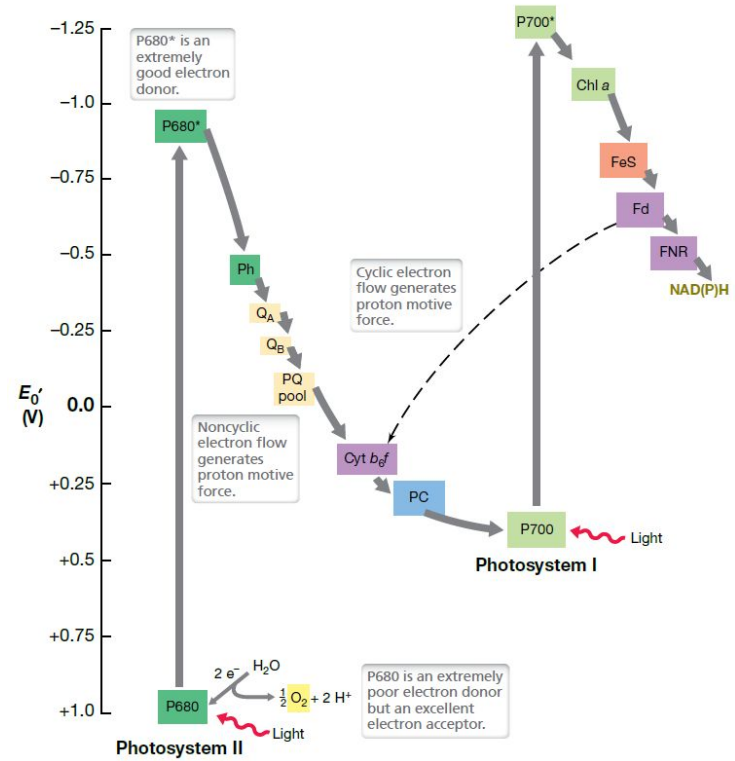
Carotenoids are hydrophobic pigments that absorb light in the blue region of the spectrum. They function primarily as **photoprotective agents**, absorbing harmful bright light (UV). This **prevents** the **catalysis of photooxidation reactions** and the subsequent **formation of toxic oxygen species**.

Phycobilins, mainly divided in phycocyanins and phycoerythrins, absorb light of diverse wavelengths, filling the "green gap", and transfer energy to the reaction center chlorophyll.

Oxygenic Photosynthesis

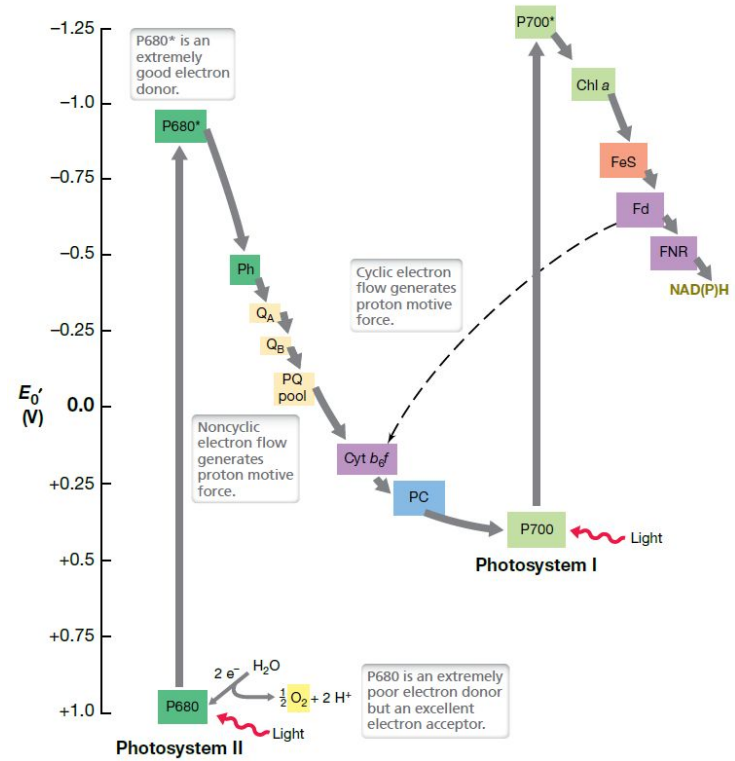
Electrons travel through an **electron transport chain** whose components are arranged in the membrane based on their reduction potential (E_0').

Electron flow (represented in the "Z" scheme) starts with the photosystem II (PSII) in which the **P680 chlorophyll(a)** receives excitation energy from light, switching to an excited form $P680^*$ that become a strong electron donor. The loosened electron of $P680^*$ is taken up by the **primary electron acceptor** of the chain, a **pheophytin**. $P680^*$ is **oxidized** to a cationic form $P680^+$, a strong electron acceptor, **responsible** of the **water oxidation**. The oxidation of water by PSII occurs at the **oxygen-evolving complex** and is catalyzed by a Mn_4Ca cluster. $P680^+$ removes electrons from the Mn_4Ca cluster, which, in turn, removes electrons from H_2O molecules, resulting in the production of O_2 and H^+ .

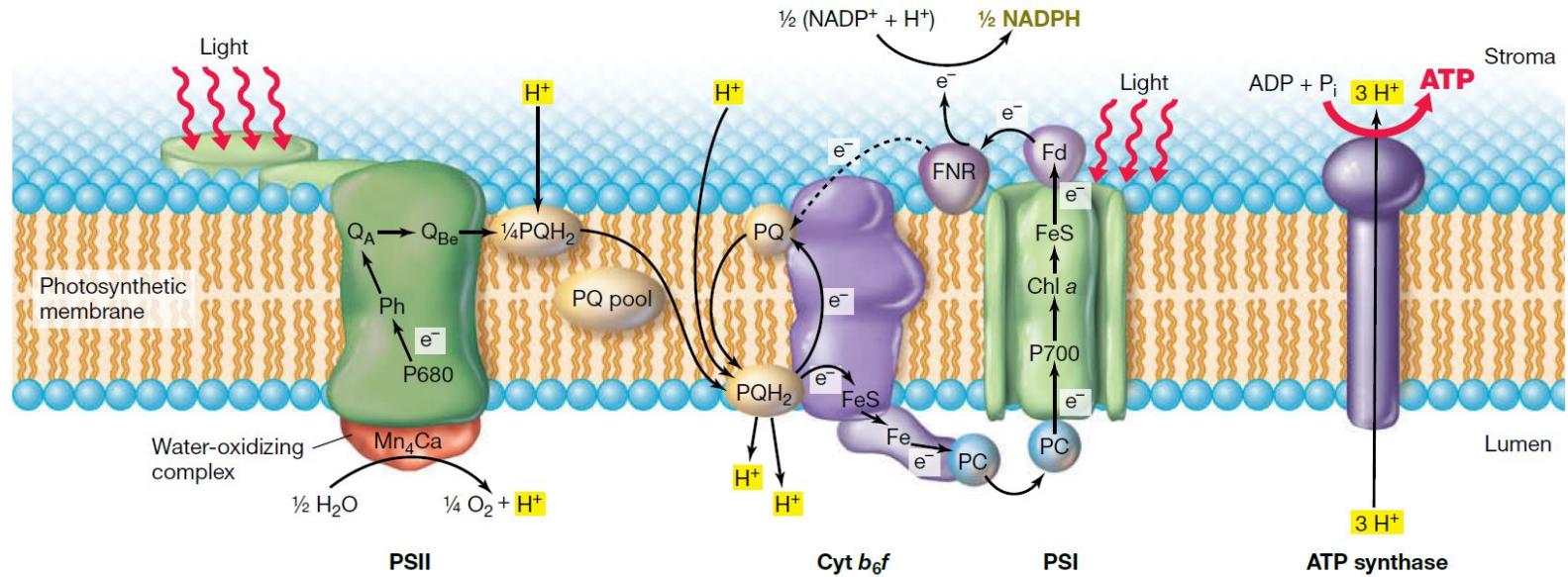


Oxygenic Photosynthesis

Each electron transferred to pheophytin travels through the electron transport chain, generating the **proton motive force**. Electrons from plastoquinone are transferred through cytochrome b_6f , containing a FeS cluster, and through a copper-containing plastocyanin before being donated to the PSI reaction center. Here, chlorophyll P700 absorb light and continue the electron transport chain, leading to the **final reduction of NAD(P)⁺ to NAD(P)H** (reducing power necessary for CO₂ fixation) by means of a ferredoxin molecule. Oxygenic photosynthesis results in **noncyclic photophosphorylation**. However, when the cell requires more ATP, electrons cycle back in the electron transport chain, generating proton motive force and ATP synthesis, in a process called **cyclic photophosphorylation**.



Oxygenic Photosynthesis



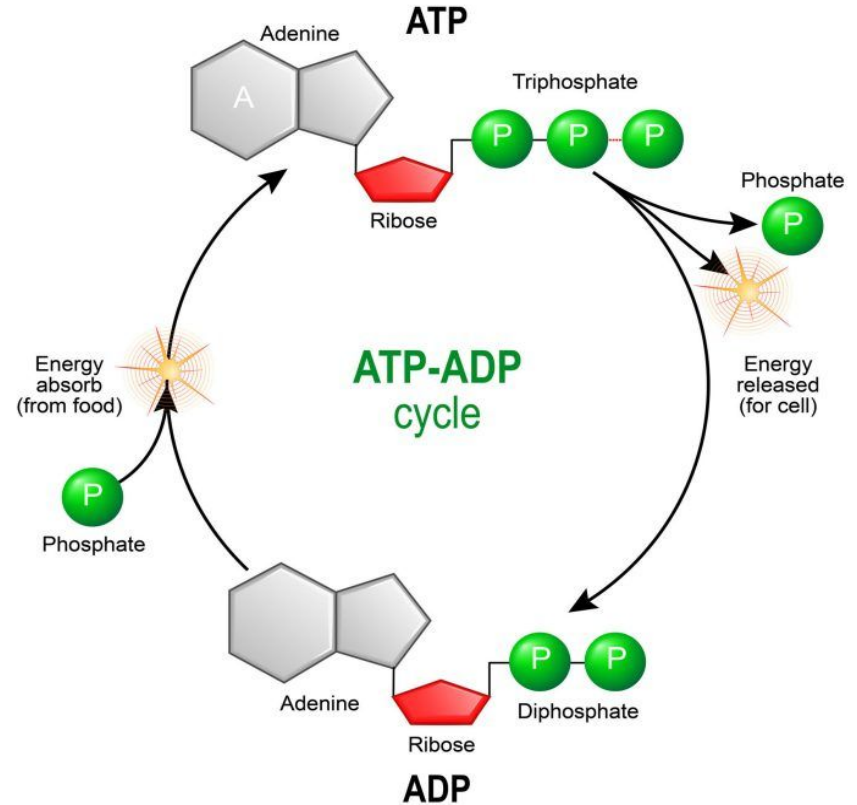
2 protons are generated for 1 H_2O molecule that is split by PSII.

Other 4 protons are translocated across the membrane every 2 electrons transferred through the electron transport chain, resulting in a total of 12 protons translocated for every molecule of O_2 produced.

2 electrons are used to obtain 1 NADPH molecules, with a $H_2O/NADPH$ ratio of 1. 3 protons are used by ATP synthase to produce ATP, in a H_2O/ATP ratio of 2/3

Oxygenic Photosynthesis

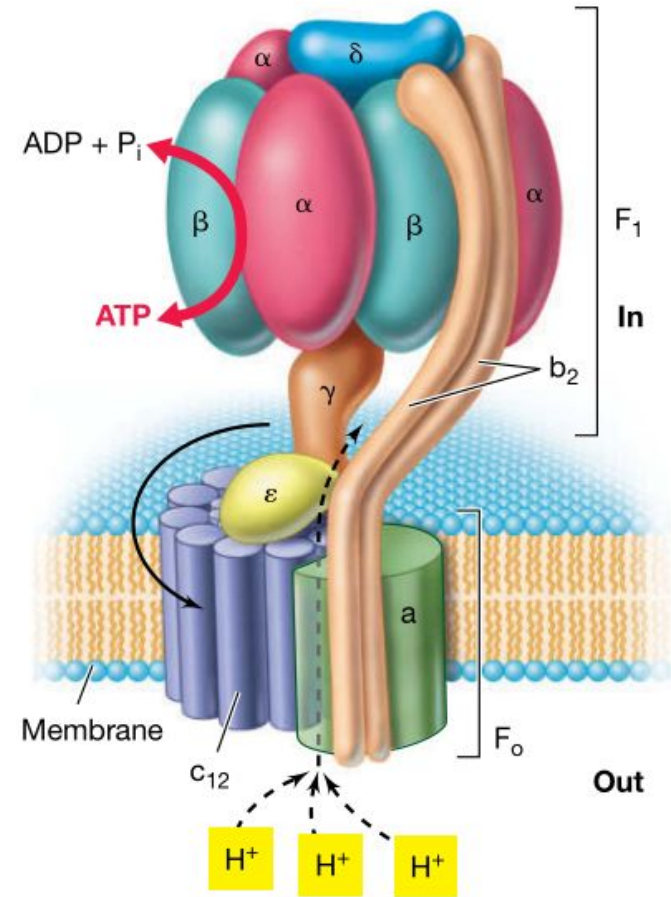
Adenosine triphosphate or ATP is a molecule composed of adenine (a nitrogenous base), ribose (a 5-carbon sugar), and three phosphate groups. It works as the primary energy storage in the cell. ATP stores this energy in its **high-energy phosphate bonds**. When these bonds are broken (particularly the bond between the second and third phosphate), energy is released to drive various cellular processes.



ATP Synthase (ATPase)

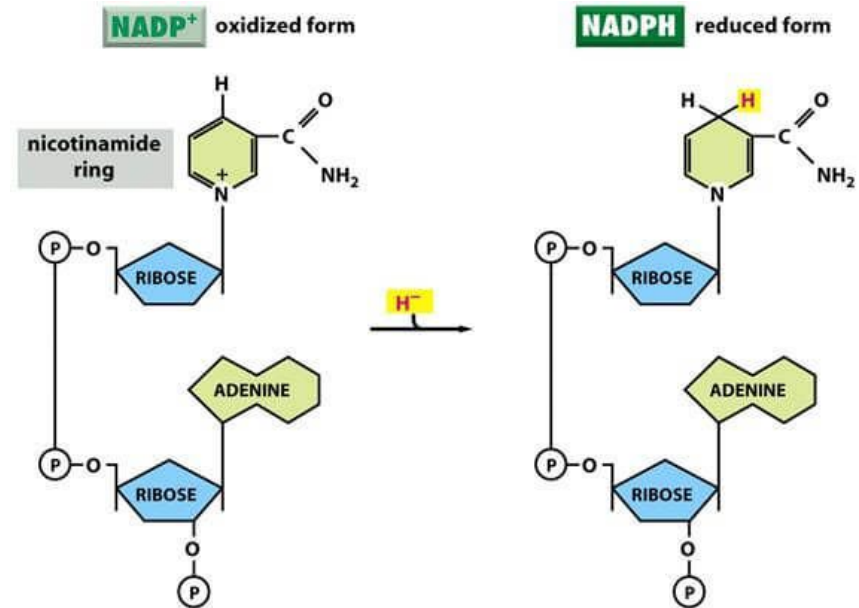
ATPase works through a mechanism called chemiosmosis, which is driven by the proton motive force (PMF). Structurally, it is composed by a transmembrane F_0 Region, which acts as a proton channel, and a cytoplasmic F_1 Region, composed of several subunits that form a rotating motor containing the catalytic sites where ATP synthesis occurs.

During photosynthesis or respiration, protons accumulate outside the membrane, creating an electrochemical gradient, or proton motive force (PMF). To dissipate this force, the F_0 Region allows protons to flow down their gradient. The movement of the protons generates energy, which causes a rotational motion in the ATPase complex. This rotation drives conformational changes in the F_1 Region, where the subunits recognize and bind a molecule of ADP and a phosphate group. When the F_1 Region returns to its original conformation, the energy released is used to catalyse the ATP synthesis.



NADP(H) And Reducing Power

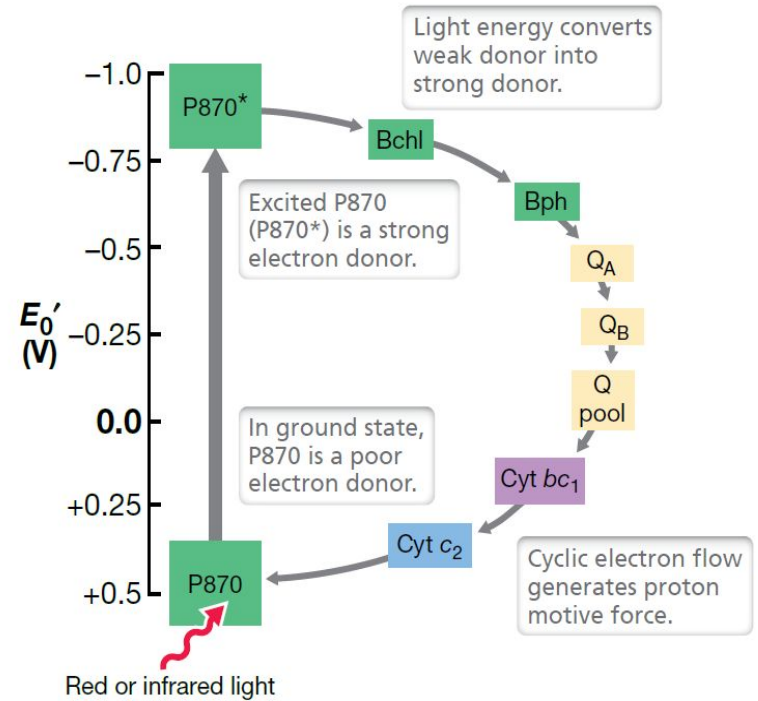
Nicotinamide adenine dinucleotide phosphate or NADP(H) is a critical molecule in microbial cells, playing a key role in anabolic reactions, which are biosynthetic processes that build cellular components. During oxygenic photosynthesis, the oxidised form NADP^+ acts as the final electron acceptor in the light reactions, completing the transfer of energy from sunlight into a chemical form. The reduction of NADP^+ to NADPH is crucial because NADPH is a fundamental source of reducing power, due to its strong tendency to donate electrons. In fact, during the Calvin cycle, it helps reduce carbon dioxide (CO_2) into organic compounds (glucose) providing the necessary electrons



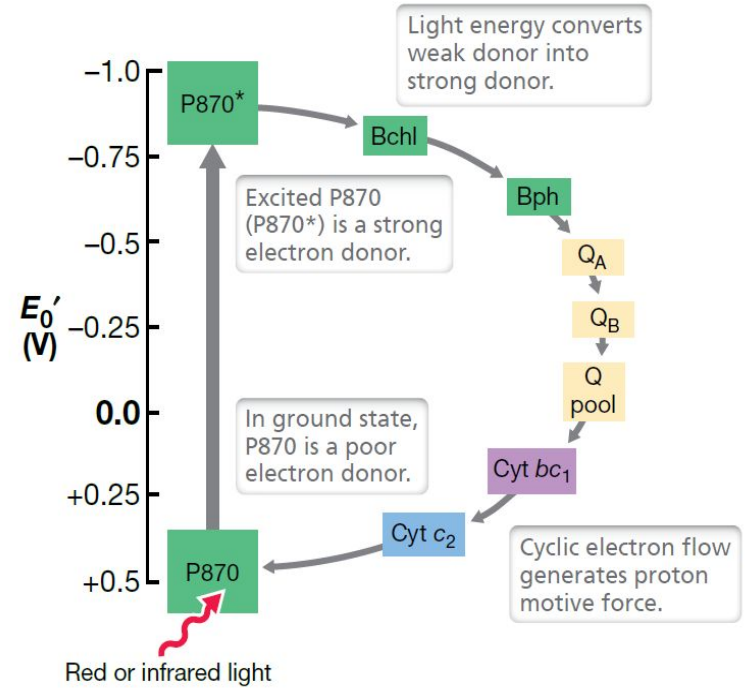
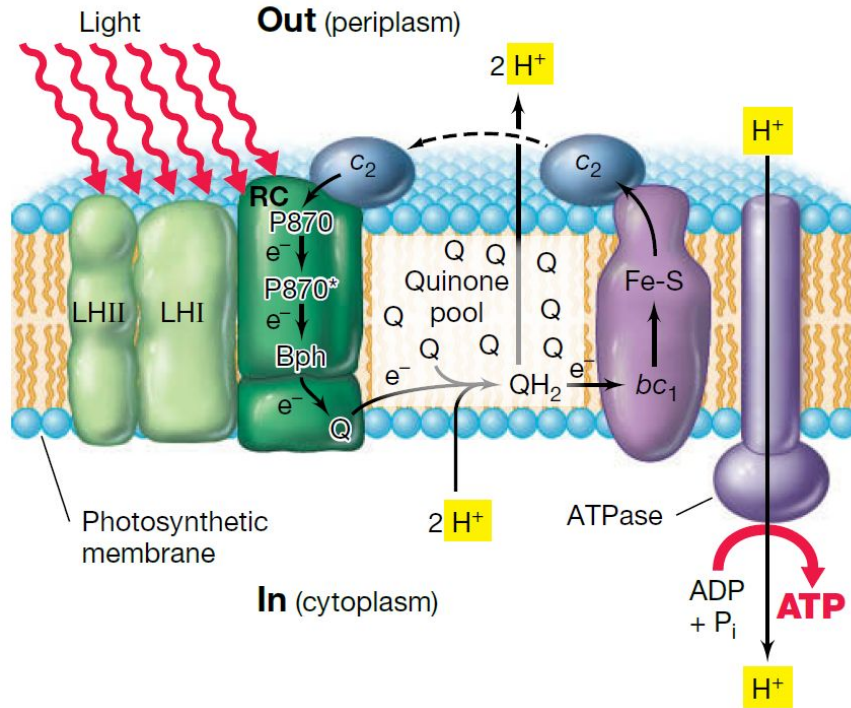
Anoxygenic Photosynthesis

Anoxygenic phototrophs only have one photocomplex, where the “special pair” of bacteriochlorophyll(a) P870 starts the process absorbing light energy. Once P870* absorbed a photon, it ejects an electron, which is transferred through another molecule of bacteriochlorophyll to a bacteriopheophytin(a). This initial charge separation yields a positive charge on P870+ that will be, at the end of the cyclic electron flow from a cytochrome c2, which, reducing P870+, acts as an electron-shuttle. Electron flow generates proton motive force, thus leading to the synthesis of ATP. Unlike oxygenic phototrophs, anoxygenic photosynthetic organisms do not use H₂O as electron donor, but generally reduced sulfur compounds such as H₂S.

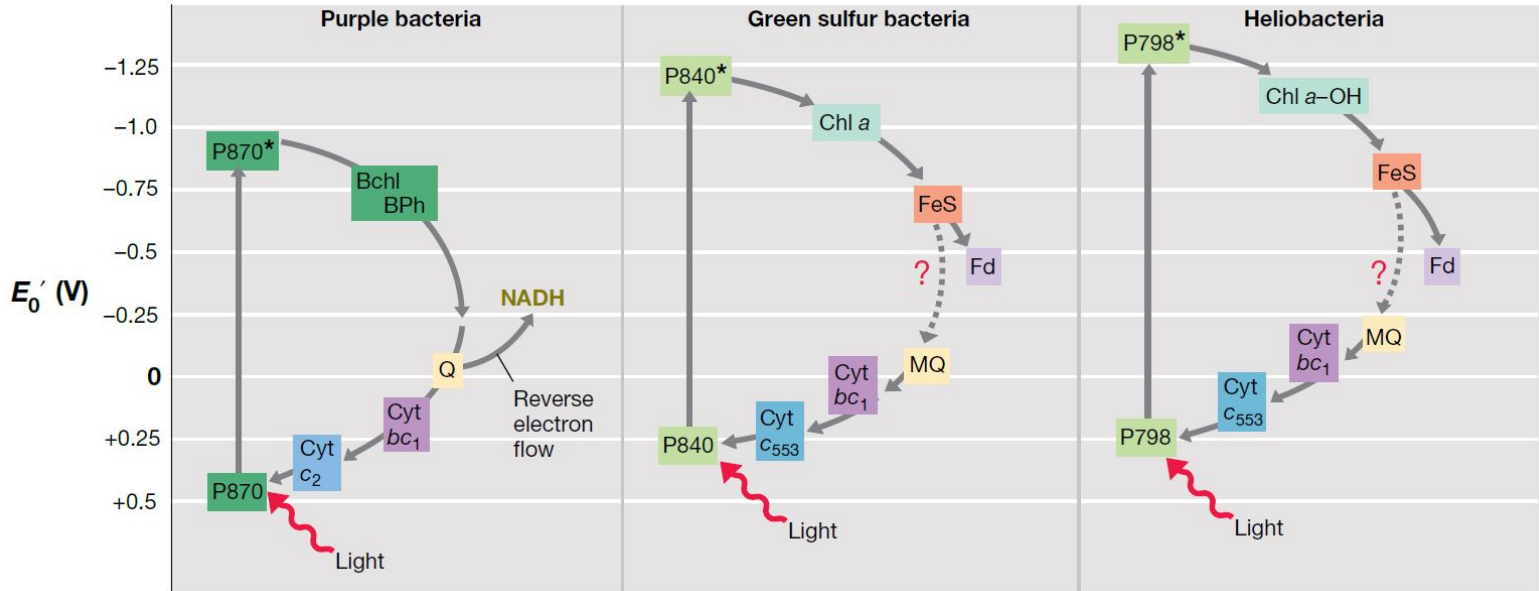
This generates the reducing power necessary for NAD(P)⁺ reduction to NAD(P)H, which is in turn required for carbon fixation.



Anoxygenic Photosynthesis



Anoxygenic Photosynthesis

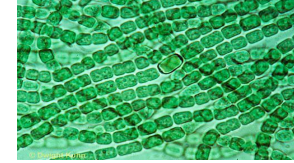


In purple sulfur bacteria, electrons coming from the oxidation of H_2S are transferred from bacteriochlorophylls to a quinone. E_0' of quinone is insufficiently electronegative to reduce NAD^+ , therefore electrons must be forced backwards (against the electrochemical gradient). This energy-requiring process, called reverse electron transport, is driven by the energy of the proton motive force. In green sulfur bacteria and Heliobacteria remains unclear whether electron transfer is cyclic or noncyclic

Anoxygenic Photosynthesis

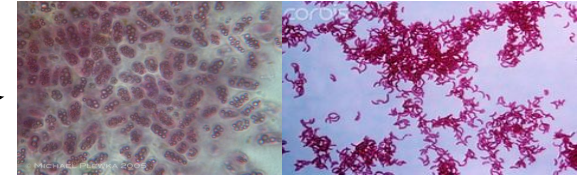
Prokaryotic Oxygenic Photoautotrophs:

Cyanobacteria
(formerly known also as Cyanophyta or Blue-Green Algae)



Prokaryotic Anoxygenic Photoautotrophs:

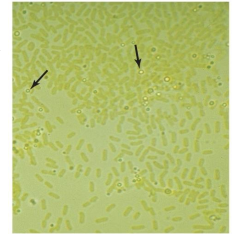
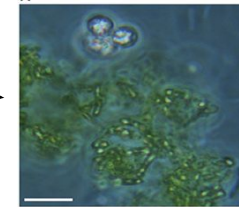
Purple Bacteria (Chromatiaceae and Rhodospirillaceae)



Green Sulfur Bacteria (Chlorobi)

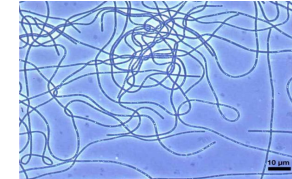
Phototrophic Acidobacteria (Chloracidobacterium)

Phototrophic Heliobacteria (Heliobacteriaceae)



Prokaryotic Photoheterotrophs:

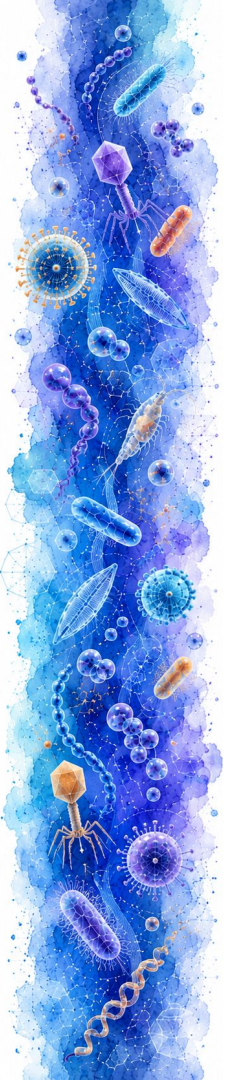
Green non-sulfur bacteria
(Chloroflexi)



**ME EXPLAINING
BACTERIAL METABOLISM
AND BIOSYNTHETIC PATHWAYS**

LITERALLY EVERYONE ELSE





This Week Read

Sattley, W.M., Madigan, M.T. (2014). The Family Heliobacteriaceae . In: Rosenberg, E., DeLong, E.F., Lory, S., Stackebrandt, E., Thompson, F. (eds) The Prokaryotes. Springer, Berlin, Heidelberg.
https://doi.org/10.1007/978-3-642-30120-9_362

Deep-Sea Exploration Streaming:

Schmidt Ocean Institute

<https://www.youtube.com/@SchmidtOcean>

Romanche Fracture Zone West Station 1 | SOI Divestream 969 Part 2

<https://www.youtube.com/watch?v=L9dQqpf9B6g>